



TRAINER SLIDES · WSQ

AI Security for Autonomous AI Agents

WSQ Course Code: TGS-2025060473 · 2 Days · 16
Hours

Skills Framework TSC: Generative AI Principles and Applications
(ICT-INT-0052-1.1)

Conducted by Tertiary Infotech Pte Ltd · UEN 20120096W

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WSQ

AI SECURITY



Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer or administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.
- Complete the TRAQOM survey at the end of the course — it is required for funding.

About the Trainer



Your Trainer

Complete this profile before delivery

NAME

QUALIFICATIONS

INDUSTRY EXPERIENCE

TRAINING EXPERIENCE

About the Trainer



Dr Alfred Ang

Principal Trainer
Tertiary Infotech Pte Ltd

ROLE

Principal Trainer

EXPERTISE

AI security, governance and enterprise IT

Who Is in the Room?

1

Role

What systems do you own or use?

2

Exposure

What data and tools can they reach?

3

Concern

What outcome would be hardest to reverse?

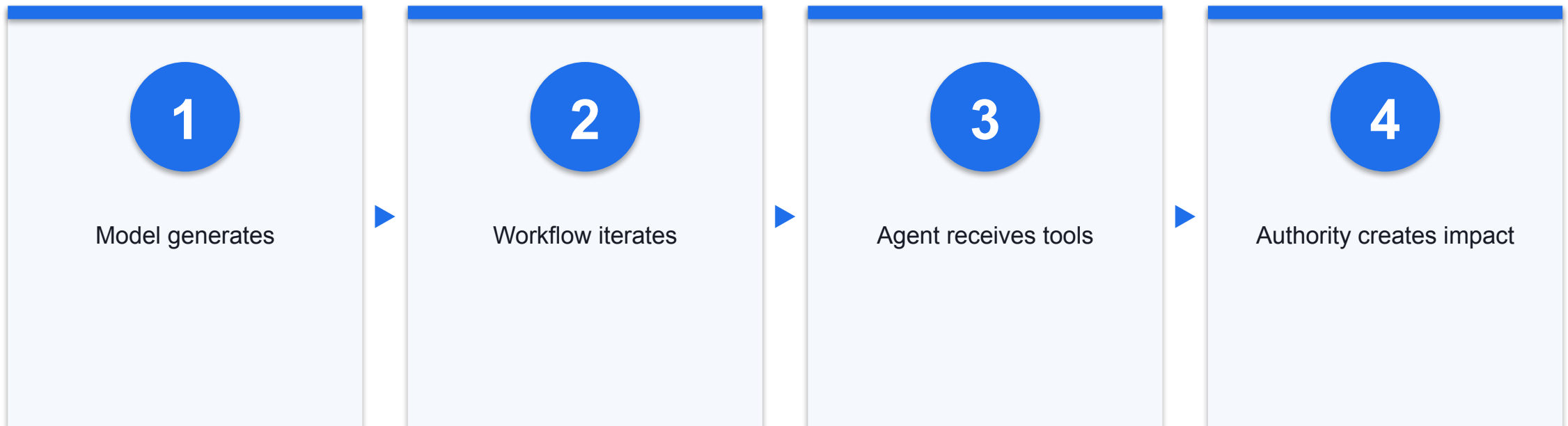
Working Agreements for Security Labs

- Synthetic data only
- Local dummy tools
- Share evidence, not secrets

- No live credentials
- No real recipients
- No production probing

Evidence: SYN

Why This Course Exists



Security changes when probabilistic output can trigger actions in real systems.

Two-Day Learning Journey



Learning Outcomes

1

LO1

Explain GenAI concepts and applications

2

LO2

Test prompt variations and controls

3

LO3

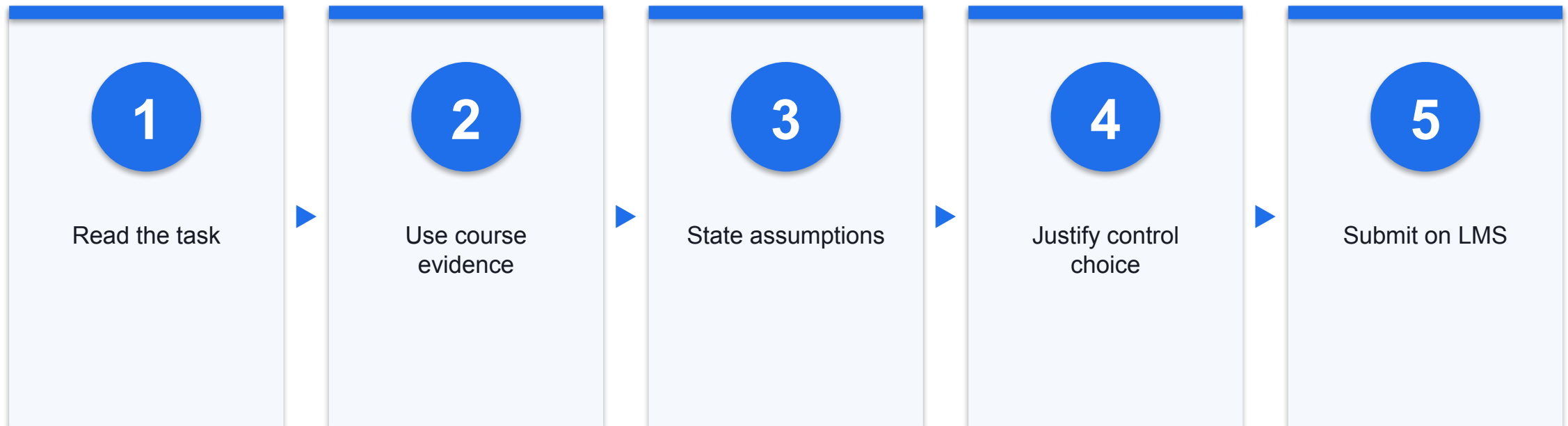
Analyse ethical, bias and security risks

K1–K5 and A1–A5 Evidence Map

Evidence family	Statements	Where demonstrated
Models and data	K1–K3, K5	History, taxonomy, poisoning
Prompts and metrics	K4, A5	Injection activity
Use, limits, ethics	A1–A4	Cases and deployment gate

Evidence: ADMIN

Briefing for Assessment

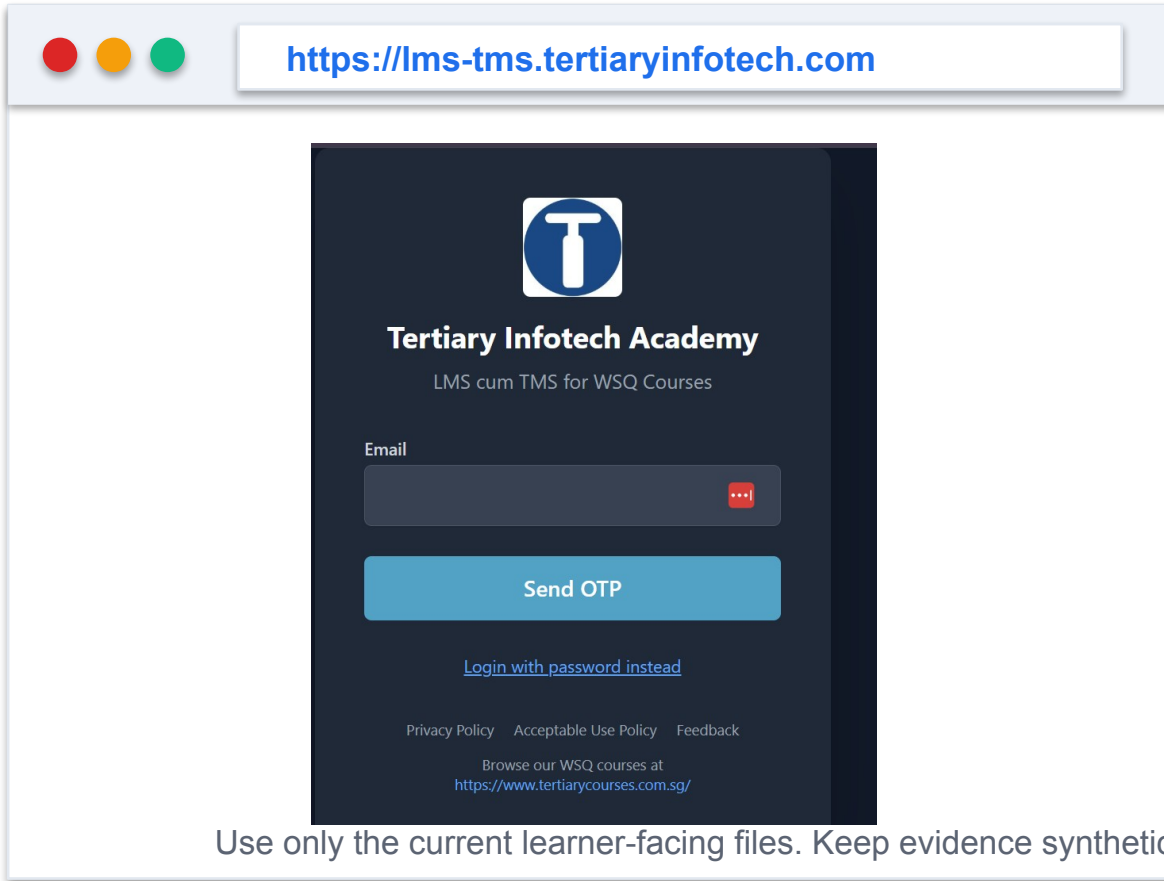


Assessment Components and Boundaries

- Written Assessment
- Knowledge statements K1–K5
- Open book

- Case Study
- Learning outcomes and abilities
- Evidence-based recommendation

Courseware, Activities, and Safe Lab Environment



The screenshot shows a web browser window with the address bar displaying <https://lms-tms.tertiaryinfotech.com>. The page content is a dark-themed login form for 'Tertiary Infotech Academy'. At the top is a logo consisting of a white 'T' inside a blue circle. Below the logo, the text 'Tertiary Infotech Academy' is displayed, followed by 'LMS cum TMS for WSQ Courses'. There is an 'Email' input field with a red eye icon for toggling visibility. Below the input field is a blue 'Send OTP' button. A link 'Login with password instead' is positioned below the button. At the bottom of the form, there are links for 'Privacy Policy', 'Acceptable Use Policy', and 'Feedback', followed by the text 'Browse our WSQ courses at <https://www.tertiarycourses.com.sg/>'.

Use only the current learner-facing files. Keep evidence synthetic, local and reversible; never upload secrets or answer keys.

1

Step 1

Download current learner-facing files from the LMS

2

Step 2

Use the five activity folders as the evidence workspace

3

Step 3

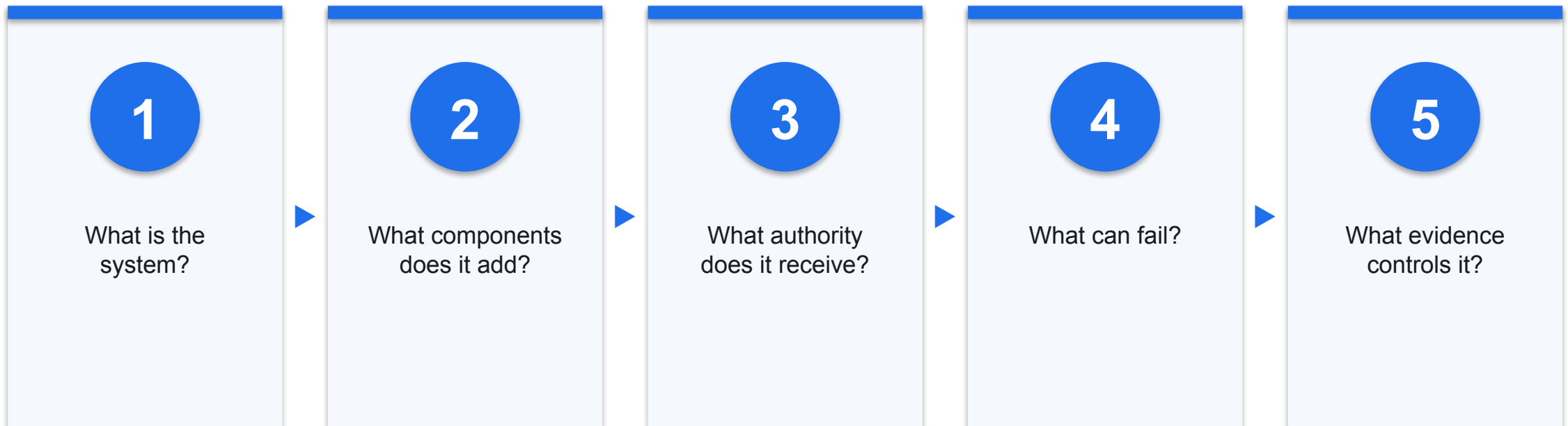
Run only synthetic, local and reversible tests

COURSE MODULE

Day 1 — From AI Models to Acting Systems

Origins → generation → agency → product anatomy → threat model

The Dependency Map: What Must Be Understood First



Evidence: SYN · Sources: S09, S12, S14

Before 'Generative': What Did AI Mean?

1

AI system

Engineered system producing predictions, recommendations, decisions or content

2

Objective

Outputs influence virtual or physical environments

3

Autonomy

Systems operate with varying levels of autonomy

1950 — The Machine-Intelligence Question

- Turing reframed the question through observable conversational behaviour
- The paper discussed learning machines as well as fixed programs
- It predates the term artificial intelligence

1955–1956 — ‘Artificial Intelligence’ Is Named

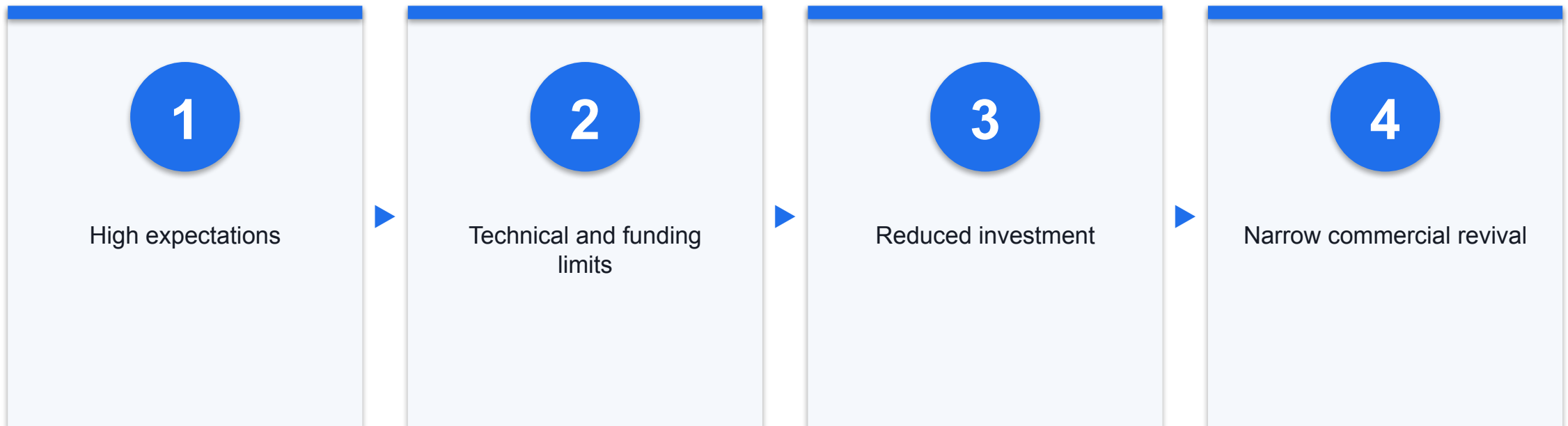
- The Dartmouth proposal used the term artificial intelligence
- The summer project took place in 1956
- The proposal treated learning and intelligence as problems that could be described precisely

Symbolic AI and Early Learning Systems

- Symbolic systems
- Represent knowledge as rules and symbols
- Reason through explicit operations

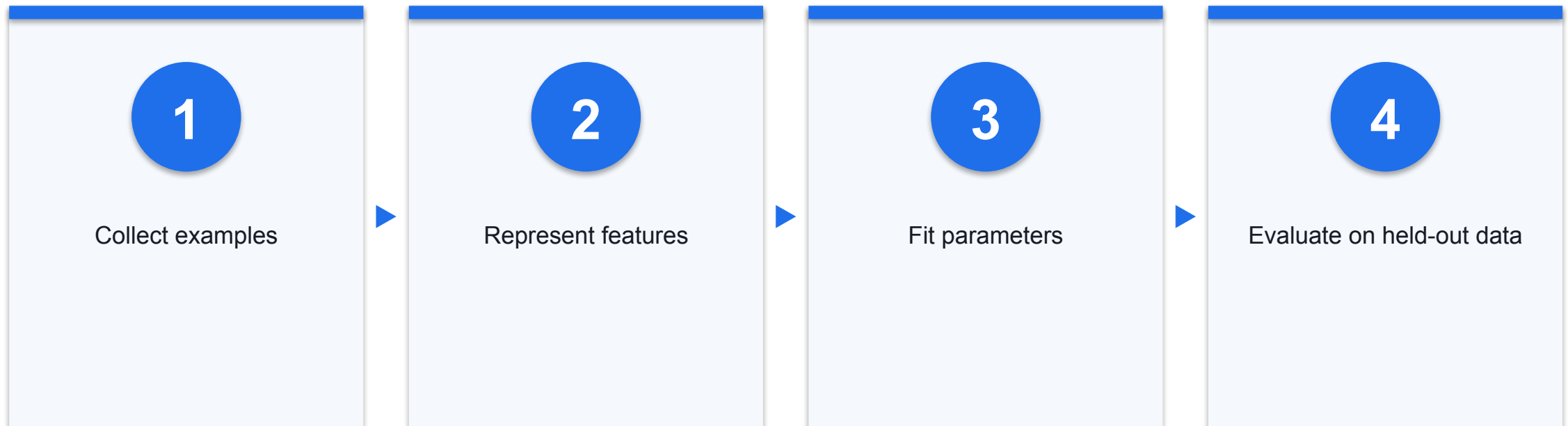
- Learning systems
- Adjust behaviour from examples or feedback
- Depend on data and objective design

Cycles of AI Expectation and Investment



The history is cyclical; avoid a single-cause story for either progress or slowdown.

Statistical Machine Learning — Course Synthesis



The decision boundary is learned from data rather than authored as a complete rule set.

2012 — Deep Learning's Visibility Breakthrough

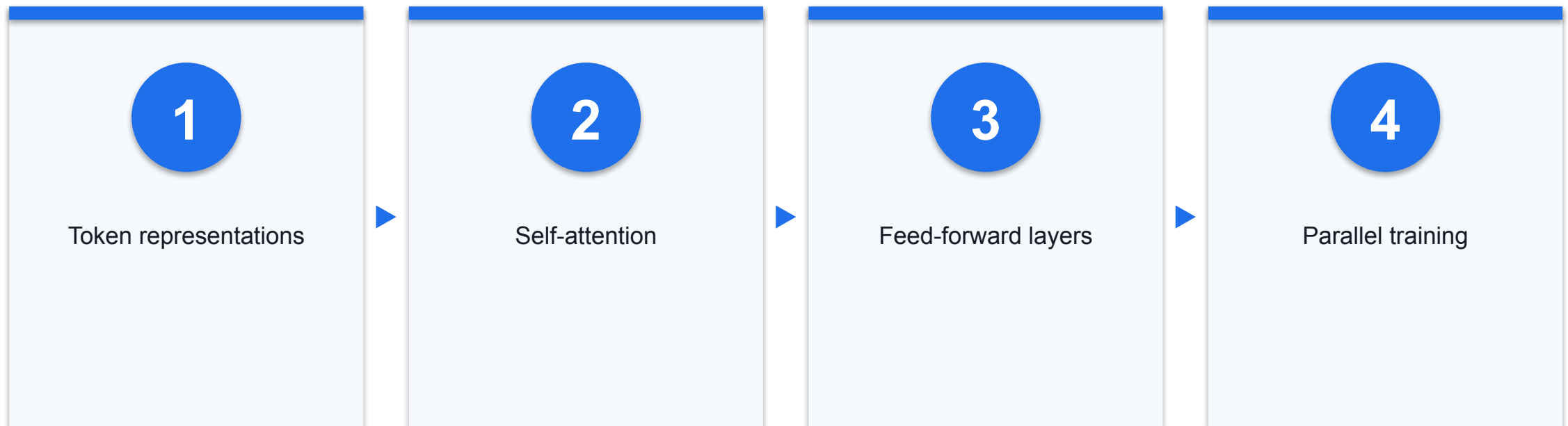
- AlexNet used a deep convolutional network for ImageNet classification
- The published system used GPUs to train the model
- Its result helped demonstrate the practical impact of deep learned representations

Sequence Models and the Attention Problem

- Many earlier neural sequence models
- Recurrence processes tokens in order
- Long dependencies are difficult

- Attention
- Weights relationships between positions
- Creates shorter paths between relevant tokens

2017 — The Transformer



The original architecture dispensed with recurrence and convolution for its sequence-transduction tasks.

From Large Models to Foundation Models

1

Pre-train

Learn broad patterns from large, diverse corpora

2

Adapt

Prompt, fine-tune or retrieve for a task

3

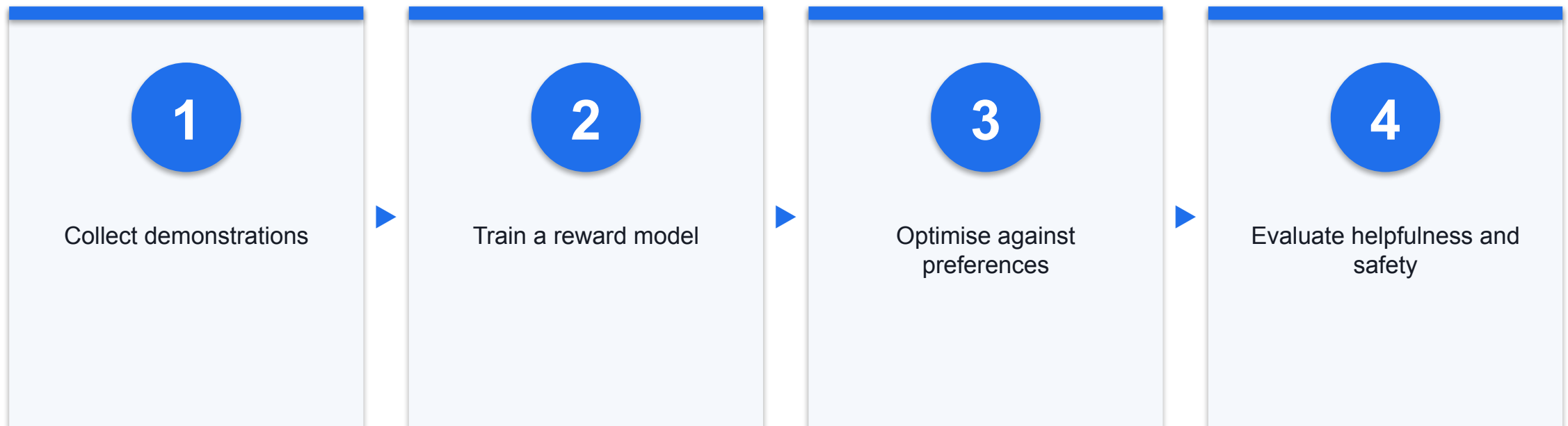
Reuse

One model supports many downstream applications

2020 — Few-Shot Behaviour at Scale

- GPT-3 was evaluated on tasks described through text prompts
- Few-shot examples were provided in context rather than through task-specific gradient updates
- Performance varied substantially by task and prompt

Instruction Following and Human Feedback

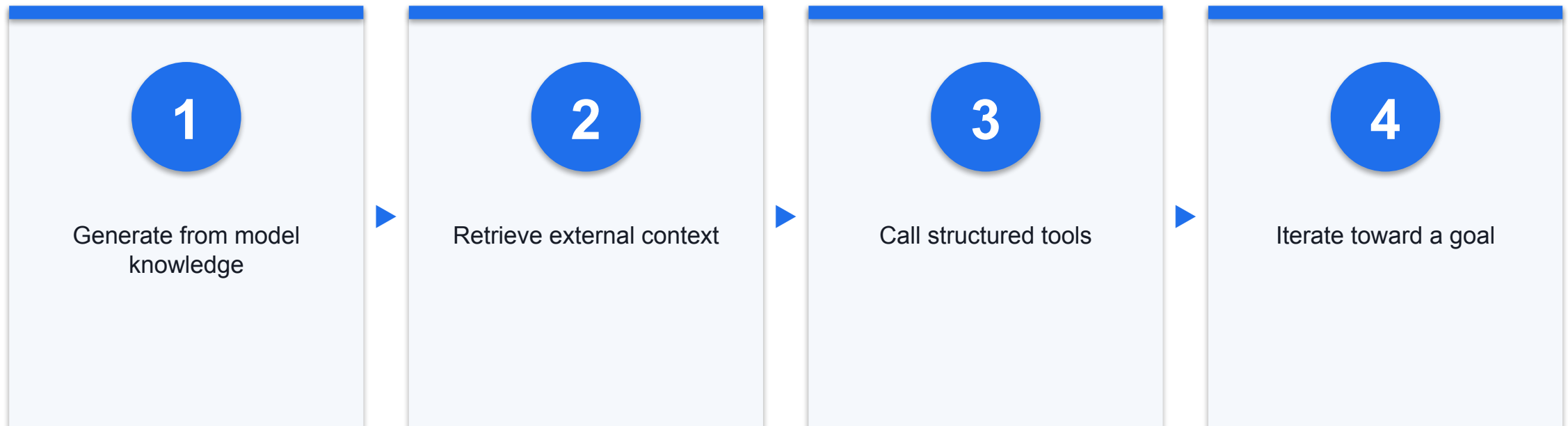


Human feedback changes behaviour; it does not create a deterministic policy boundary.

November 2022 — ChatGPT Broadens the Conversational Interface

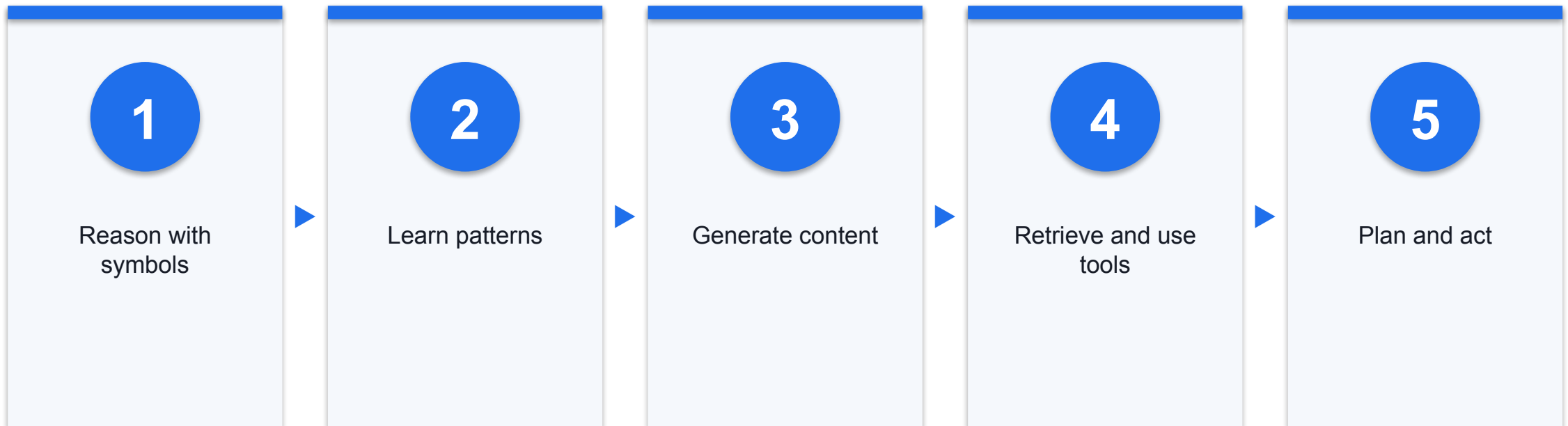
- ChatGPT's public release in November 2022 made multi-turn language-model interaction widely visible
- Natural-language prompts exposed many tasks through one interface
- A conversational interface did not remove limitations, variability or data risk

Retrieval and Tool Use Move Beyond Chat



Each step adds utility and a new trust boundary.

The Evolution in One View



This is an instructional synthesis, not a claim that one paradigm replaced every earlier one.



Morning Break

KEY IDEA

A Taxonomy Warning: The Terms Are Not Standardised

Sources use overlapping labels · The course declares operational definitions · Classify the architecture, not the marketing name

Generative AI — Course Definition

1

Primary purpose

Generate synthetic text, image, audio, video or code

2

Interaction

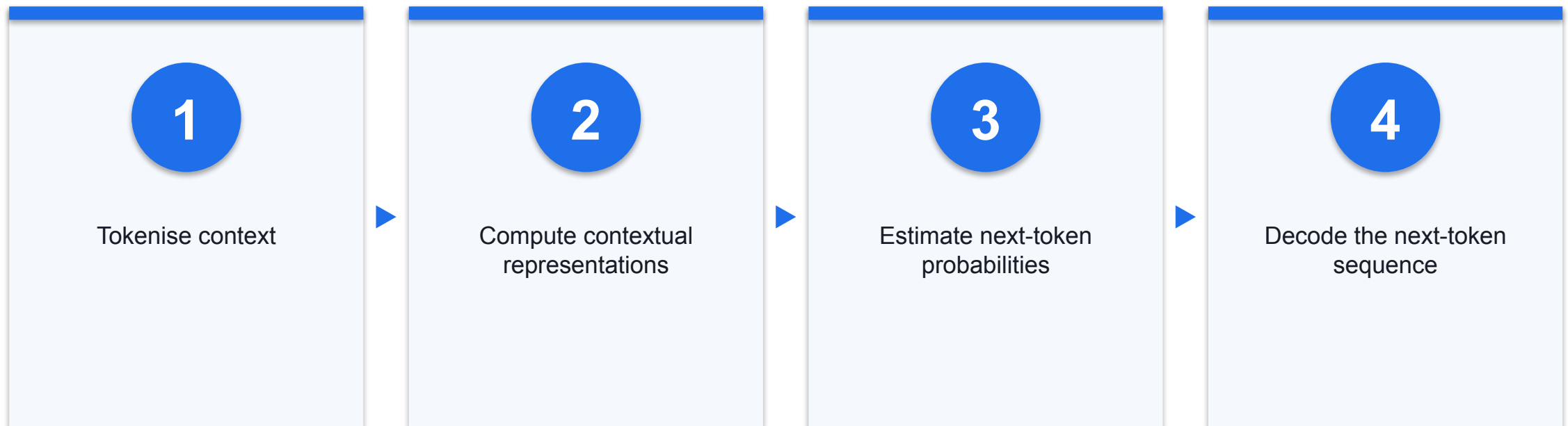
Respond to supplied context and prompts

3

Boundary

A model response alone does not imply autonomous action

How an Autoregressive Language Model Produces Text



Sampling and context make output probabilistic rather than a fixed database lookup.

Generative vs Discriminative Models — Course Comparison

- Generative
- Learns patterns used to produce new content
- May be exposed to injection, leakage, poisoning and unsafe output

- Discriminative
- Learns to predict a class or score
- May be exposed to poisoning, evasion, leakage and false positives

Generative AI Strengths

1

Synthesis

Combine patterns into a new response

2

Transformation

Rewrite, translate or structure content

3

Interface

Express many tasks in natural language

Generative AI Weaknesses

1

Confabulation

Fluent output can be unsupported or wrong

2

Variability

Prompt and sampling changes alter results

3

Provenance

Output may not reveal which source supported a claim

Generative AI Security Exposure

Surface	Failure	Impact
Prompt/context	Instruction hijack	Wrong or disclosed output
Training/RAG	Poisoning	Biased or attacker-shaped answers
Output	Unsafe downstream handling	Code, data or browser impact

Evidence: DEF · Sources: S09, S11

Agentic AI — Course Definition

1

Goal-directed

Works toward an objective across steps

2

Iterative

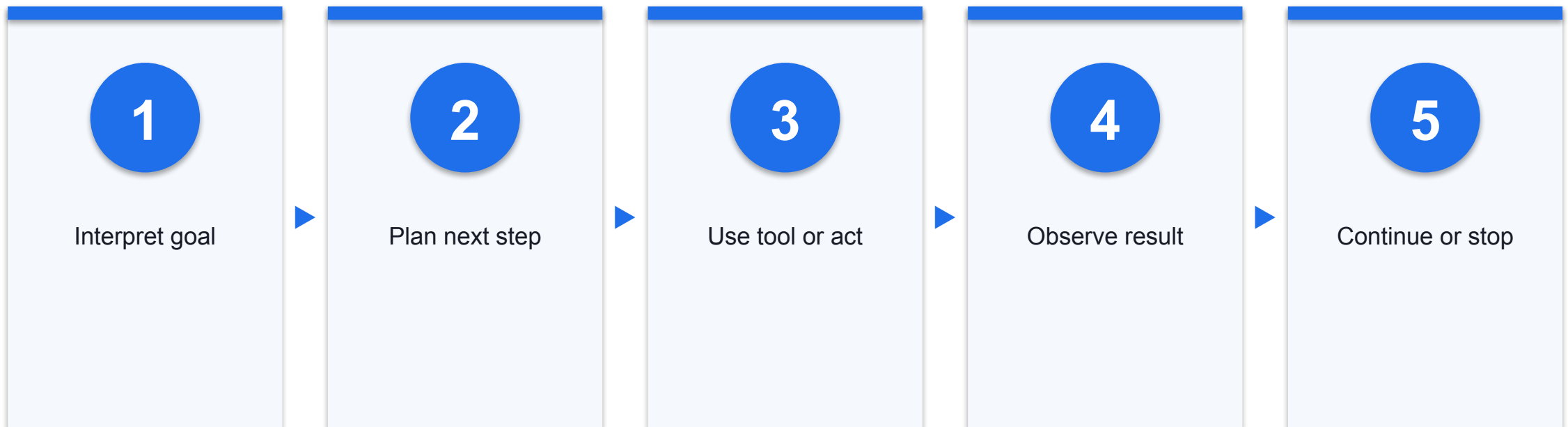
Observes results and adjusts

3

System property

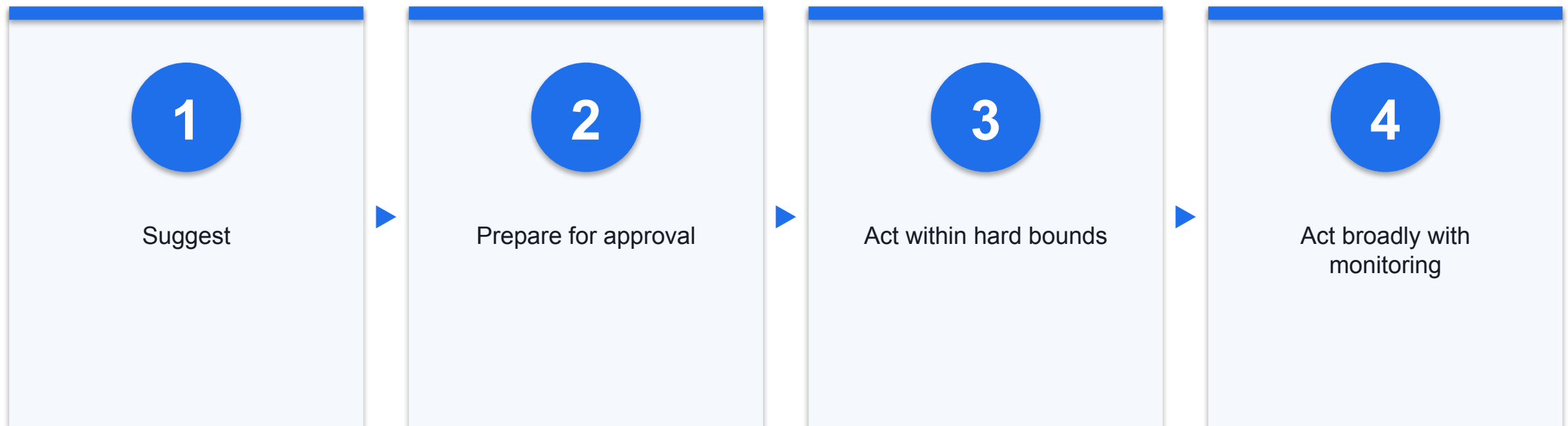
Agency comes from the model plus orchestration and authority

The Plan–Act–Observe Loop



A compromised objective can influence every later turn of the loop.

Autonomy Is a Spectrum



Risk assessment should set autonomy by impact, reversibility, data sensitivity and tool reach.

Agentic AI Strengths

1

Multi-step work

Decompose a goal into linked actions

2

Adaptation

Use observations to change the plan

3

Orchestration

Coordinate models, tools and specialised agents

Agentic AI Weaknesses

1

Compounding error

One mistake shapes later observations and plans

2

Runaway use

Loops consume time, tokens and tool capacity

3

Opaque sequence

A plausible final answer can hide a harmful path

Agentic AI Security Exposure

Agentic property	Risk	Control direction
Goal pursuit	Goal hijack	Bind goal and stop conditions
Tool choice	Tool misuse	Allowlist and validate
Delegation	Cascade/inter-agent spoofing	Authenticate and limit fan-out

Evidence: DEF · Sources: S12

AI Agent — Course Definition

1

Concrete system

A deployed software entity, not just a model

2

State and observation

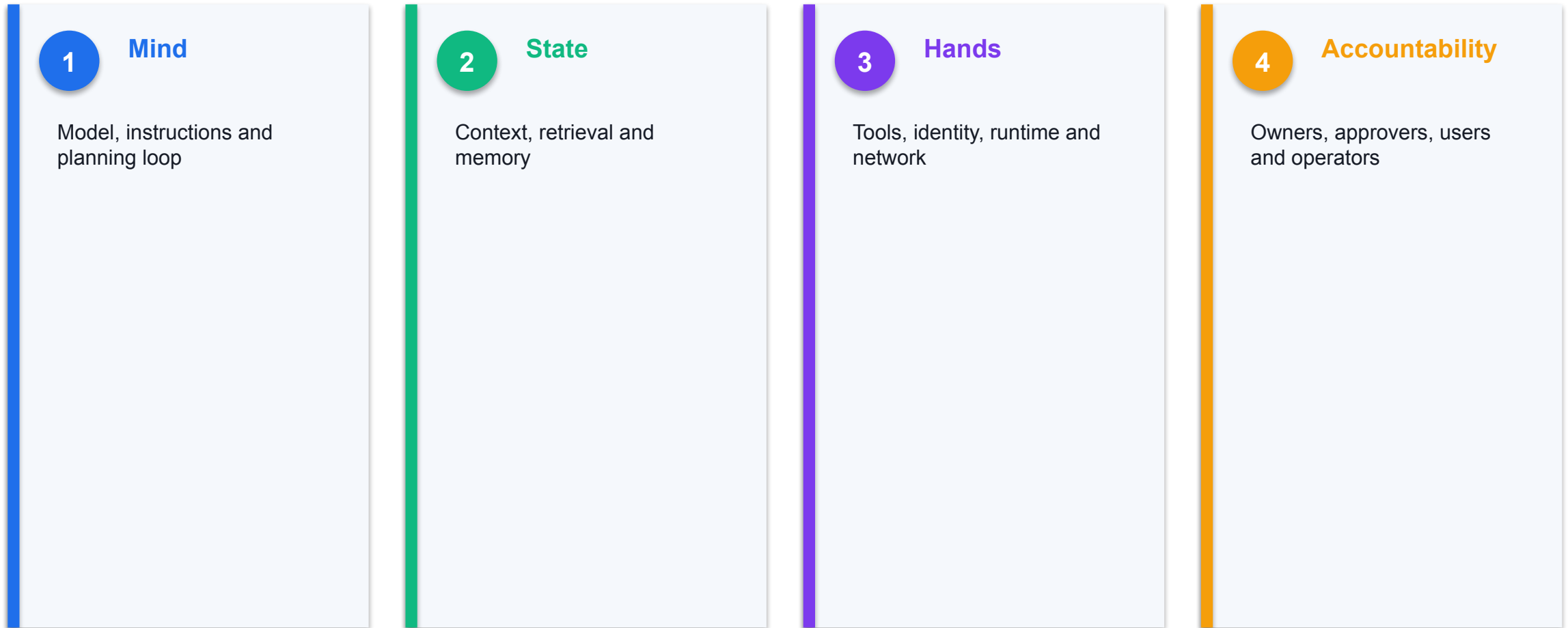
Receives environment feedback and may retain context

3

Authority

Acts through tools, identities and policies

The Agent as a Whole System



Evidence: SYN · Sources: S12

Instructions and Context

1

Trusted intent

System policy and authorised user objective

2

Untrusted content

Files, mail, web, tool output and context files

3

Core weakness

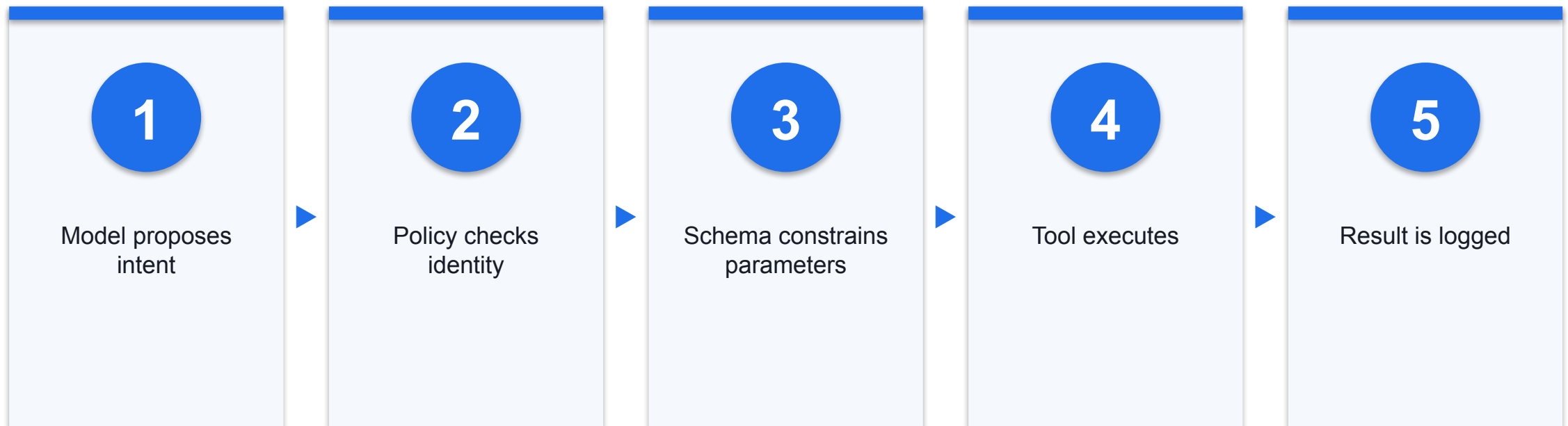
Selected or retrieved sources can become model-readable context

Retrieval and Memory

Layer	Purpose	Security concern
Prompt context	Current turn	Injection and leakage
RAG corpus	External knowledge	Poisoning and access control
Persistent memory	Cross-session state	Durable manipulation and privacy

Evidence: DEF · Sources: S09, S12

Recommended Tool-Execution Path



The identity and policy layer—not model confidence—defines authority.

Runtime, Network, and Feedback

1

Runtime

Local host, container, VM or managed service

2

Network

Inbound channels and outbound destinations

3

Feedback

Tool results and environment state become new input

AI-Agent Strengths

1

Completion

Carry a task across applications

2

Optional persistence

Where configured, resume or schedule work over time

3

Specialisation

Delegate subtasks to tools or other agents

AI-Agent Weaknesses

1

Blast radius

Compromise inherits delegated authority

2

Hidden state

Memory and intermediate actions
complicate review

3

Responsibility

Multiple providers and operators can
obscure ownership

AI-Agent Security Exposure

Axis	Exposure	Potential effect
Content	Injection/jailbreak	Changed goal or output
Authority	Token/tool misuse	Unauthorised action
Runtime	Code/network/supply chain	Host impact or exfiltration
State	Memory poisoning	Persistent behaviour change

Evidence: DEF · Sources: S11, S12

Three Overlapping Operating Layers

Dimension	Generative layer	Agentic pattern	Deployed agent
Primary role	Create content	Pursue goals iteratively	Operate as a concrete actor
State	Context-dependent	Loop state	May persist memory
Authority	None inherent	Configuration-dependent	Defined by tools and identity

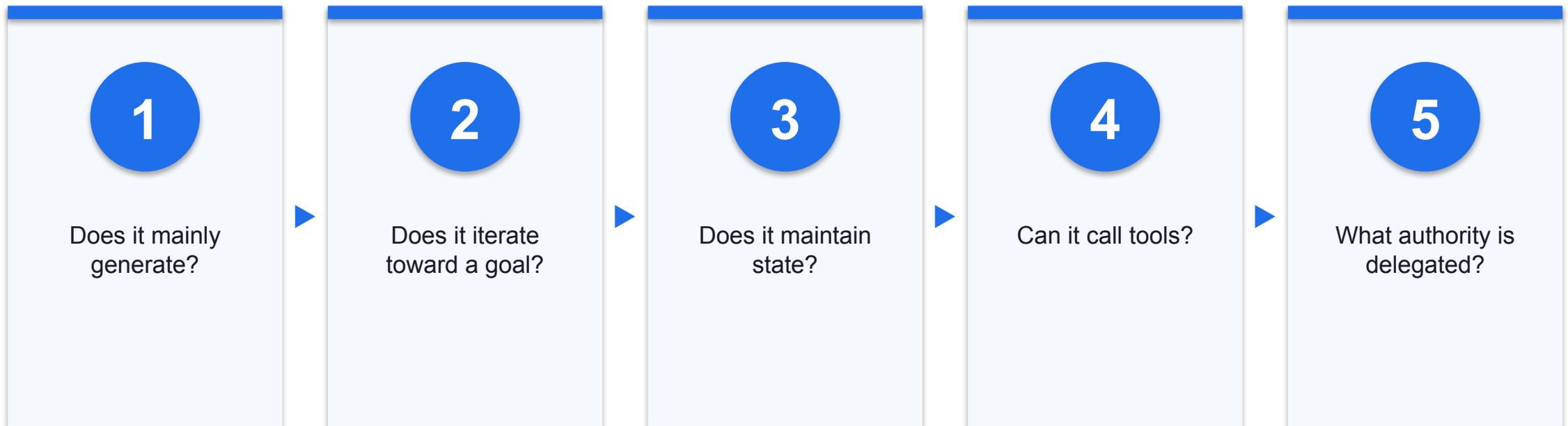
These layers can coexist in one product; they are not mutually exclusive species.

Same Model, Different Risk

- Chat configuration
- Reads one prompt
- Returns text
- Human executes any action

- Agent configuration
- Reads external sources
- Loops and calls tools
- System executes within delegated authority

Which Operating Layers Are Present?



Classify observable architecture and permissions, not the product label.

COURSE MODULE

Products Are Configurations, Not Risk Labels

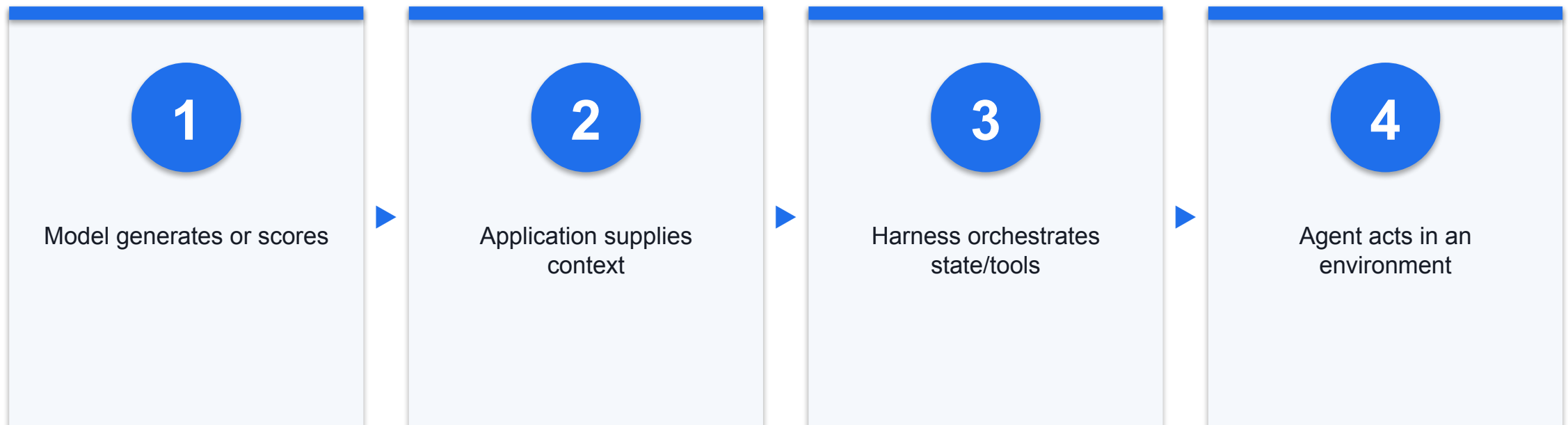
Compare capability, deployment and authority—not brand reputation.

Examples Across Three Overlapping Layers

Operating layer	Example	Decisive property
Generative	Draft or summarise on request	Returns content
Agentic	Plan, retry and route tools	Iterates toward a goal
Deployed agent	OpenClaw or Hermes deployment	Concrete runtime with tools and identity

Evidence: SYN · Sources: S09, S12

Model vs Application vs Harness vs Agent



Model choice alone does not reveal filesystem, network or credential exposure.

Four Common Deployment Patterns (May Overlap)

Deployment	Boundary	Primary review
Hosted SaaS	Provider plus tenant controls	Data terms and workspace settings
Local CLI/desktop	User account and host	Filesystem, shell and secrets
Self-hosted gateway	Organisation runtime	Network, identity and isolation
Embedded enterprise	Application and service accounts	API scopes and transaction controls

Evidence: SYN · Sources: S14

OpenClaw — Documented Architecture

1

Gateway

Connects channels, sessions and tools

2

Host state

Configuration and credentials live under the runtime state directory

3

Trust model

Documentation assumes one trusted operator boundary per gateway

As-of 20 Aug 2026; re-check before publication.

OpenClaw — Documented Security Boundaries

Boundary	Documented control	Residual concern
Senders	Allowlists and pairing	Within one trusted-operator boundary, allowed users share delegated authority
Gateway	Audit and scoped configuration	Operator credentials are sensitive
Tools	Deny high-risk tools for untrusted content	Node execution can affect a paired host

As of 20 Aug 2026; controls depend on the gateway, channel, tools and host configuration.

Hermes Agent — Documented Architecture

1

Entry points

CLI, gateway, API and library modes

2

State

Persistent memory, skills and project context

3

Authority

Built-in tools, MCP and multiple execution backends

As-of 20 Aug 2026; re-check before publication.

Hermes Agent — Documented Security Layers

Layer	Purpose	Example
Authorisation	Control who can interact	Allowlist and DM pairing
Execution	Constrain dangerous operations	Approval and container isolation
Context	Reduce malicious project instructions	Context-file scanning

As of 20 Aug 2026; permanent approvals, permissive modes and execution backend alter exposure.

Claude Code — Capability and Authority Profile

1

Local context

Reads project files within configured access

2

Tools

Terminal and MCP expand capability

3

Permissions

Allowed/disallowed tools and permission modes affect authority

As of 20 Aug 2026; sandbox and permission mode are configuration-specific. A bypass flag is not a recommended default.

Codex — Capability Depends on Surface and Mode

1

Workspace

Supported local configurations can bound writes to an authorised workspace

2

Approval

Access outside the boundary or consequential actions can require approval

3

Network

Cloud internet access is separately configurable and changes injection and exfiltration exposure

As-of 20 Aug 2026; cite the exact deployed Codex surface and settings.

ChatGPT Chat and ChatGPT Work — Current Surfaces

- Chat
- Conversational exploration and generation
- Authority depends on enabled tools and context

- Work
- Supports multi-step work with authorised files, apps and tools
- Local/cloud surface, member permissions and admin policy define the boundary

As of 20 Aug 2026; feature availability and authority depend on plan, workspace and enabled tools.

DeepSeek Model vs DeepSeek Harness

1

DeepSeek-V3 model

Model card describes weights and inference; it does not define host authority

2

DeepSeek Harness

Official repository labels dsh a developer preview with plugin-oriented architecture

3

Security decision

Treat production controls as unknown until the exact version and configuration are verified

As of 20 Aug 2026; model and harness are different layers. Do not import permissions or sandbox claims from one to the other.

Prime Agent and QM — Documented Boundaries

1

Prime Agent

Persistent coding/research harness; its documentation says the kernel is not a security sandbox

2

QM

Early organisational-agent harness with durable sandboxes, memory, skills and documented security limitations

3

Deployment gate

Verify exact version, host, command posture, egress, credentials and external containment

As of 20 Aug 2026; first-party documentation describes capabilities and limitations, not independent assurance or a safety ranking.

Eight Products — Runtime and Review Focus

Product	Documented runtime emphasis	Primary review
OpenClaw	Self-hosted gateway, channels and tools	Trusted operator, host and tool policy
Hermes	CLI/gateway/API with selectable execution backends	Approvals, backend, memory and MCP
Prime Agent	Persistent Python, skills, subagents and schedules	External isolation and user OS permissions
QM	Organisational harness with durable scoped sandboxes	Permission mode, keys and egress
DeepSeek Harness	Developer-preview plugin harness	Exact commit, plugins, listener and host
Claude Code	Repository-aware coding agent	Sandbox, terminal/MCP and permission mode
Codex	Local and cloud surfaces	Workspace, approval and network mode
ChatGPT / Work	Chat plus authorised files/apps/tools	Local/cloud surface and admin policy

As of 20 Aug 2026; documented capability is not a safety score.

OpenClaw vs Hermes — Gateway and Execution

- OpenClaw
- One trusted-operator gateway boundary
- Channels, tools, nodes and runtime state
- Sandboxing and exposure depend on configuration

- Hermes
- CLI, gateway, API and library entry points
- Memory, skills, MCP and multiple execution backends
- Approvals and backend choice change authority

As of 20 Aug 2026; compare deployments, not brand names.

Prime Agent vs QM — Persistence and Permission

- Prime Agent
- Persistent Python, subagents and schedules
- Model-generated Python and commands execute with the launching user's OS permissions
- Kernel is not a security sandbox

- QM
- Durable per-scope sandbox, memory and files
- Strict, Auto and Dangerous permission modes
- Early software with documented limitations

As of 20 Aug 2026; external containment and selected mode are part of the risk decision.

DeepSeek Harness vs Claude Code — Plugin and Repository Boundary

- DeepSeek Harness (dsh)
- Official developer preview
- Plugin-oriented harness and local UI
- Production control baseline not verified

- Claude Code
- Repository-aware coding workflow
- Terminal and MCP capability
- Sandbox and permission mode affect authority

As of 20 Aug 2026; assess the exact version and enabled extensions.

Codex vs ChatGPT Work — Workspace and Tool Authority

- Codex
- Local/cloud coding surfaces
- Workspace sandbox and approvals vary by mode
- Internet access is separately configurable

- ChatGPT Work
- Multi-step local/cloud work
- Uses authorised files, apps and tools
- Member permissions and admin policy shape effects

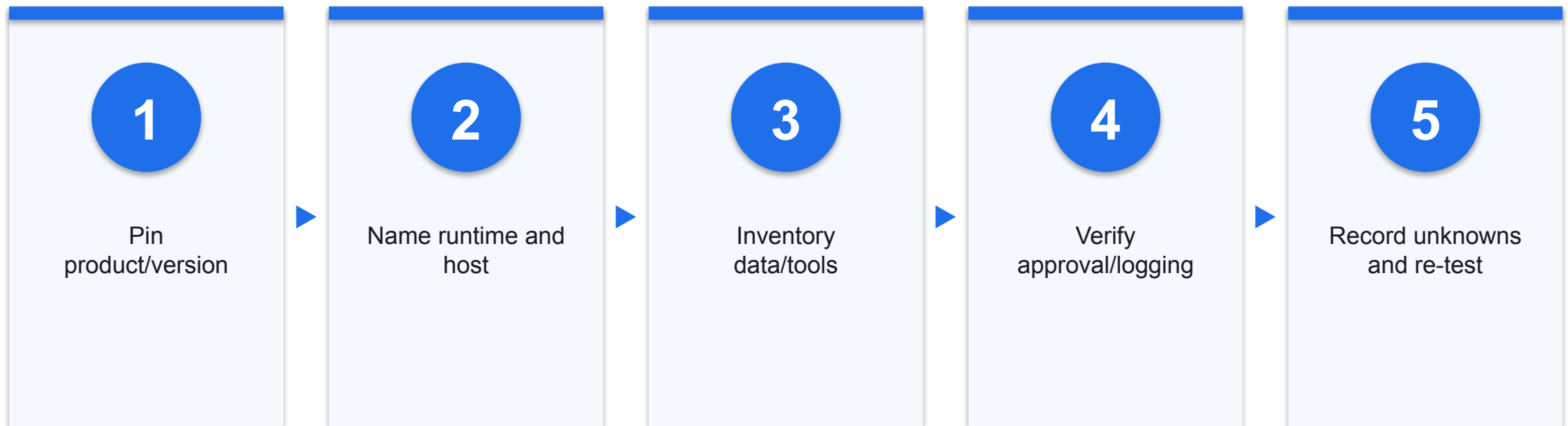
As of 20 Aug 2026; state the product surface and mode before comparing risk.

Extension and Persistence Questions Across All Eight

Axis	What to verify	Why it matters
Skills/plugins/MCP	Publisher, code, instructions, scopes and update channel	Extensions can add data and action paths
Memory/schedules	Where supported, provenance, retention and revocation	State can outlive the initiating turn
Credentials	Identity, scope, lifetime and storage	Compromise inherits delegated authority
Network	Inbound triggers and outbound destinations	Reach defines trigger and exfiltration paths

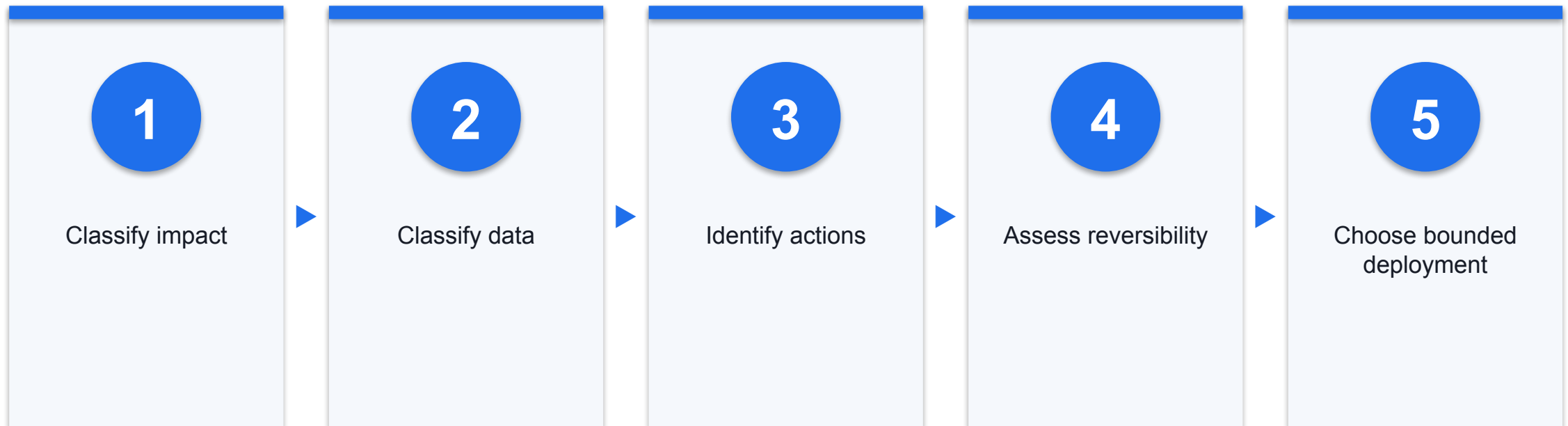
Not documented means unknown—not absent.

Product Evidence Gate Before Deployment



Re-check documentation at change or publication; no product receives a universal green/red score.

Select by Use Case and Risk Tier



A higher-capability product is acceptable only when its authority and evidence fit the risk.



Lunch Break

COURSE MODULE

Threat Modelling AI Systems

Follow content to authority, effect and evidence.

Classical Security Still Applies

1

Identity

Authenticate users, services and agents

2

Systems

Patch, segment and harden hosts

3

Operations

Log, detect, respond and recover

Evidence: DEF · Sources: S10, S13

What AI Adds or Amplifies

- Conventional
- Software vulnerabilities
- Credential theft
- Supply-chain compromise

- AI-specific/amplified
- Instruction/data confusion
- Poisoned context and memory
- Probabilistic plans driving tools

Threat Actors and Their Access Paths

1

External

Users, senders and web publishers

2

Internal

Staff, admins and data curators

3

Supply chain

Model, plugin, skill, MCP and package publishers

Assets Worth Protecting

1

Data

Personal data, prompts, files, memory and logs

2

Authority

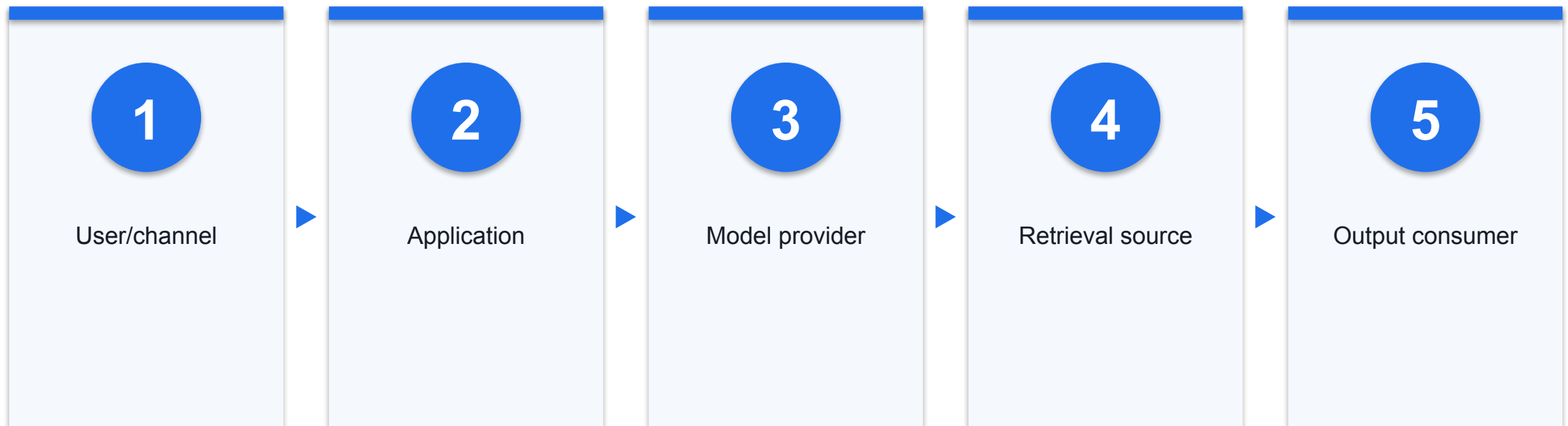
Tokens, permissions, tools and approvals

3

Decisions

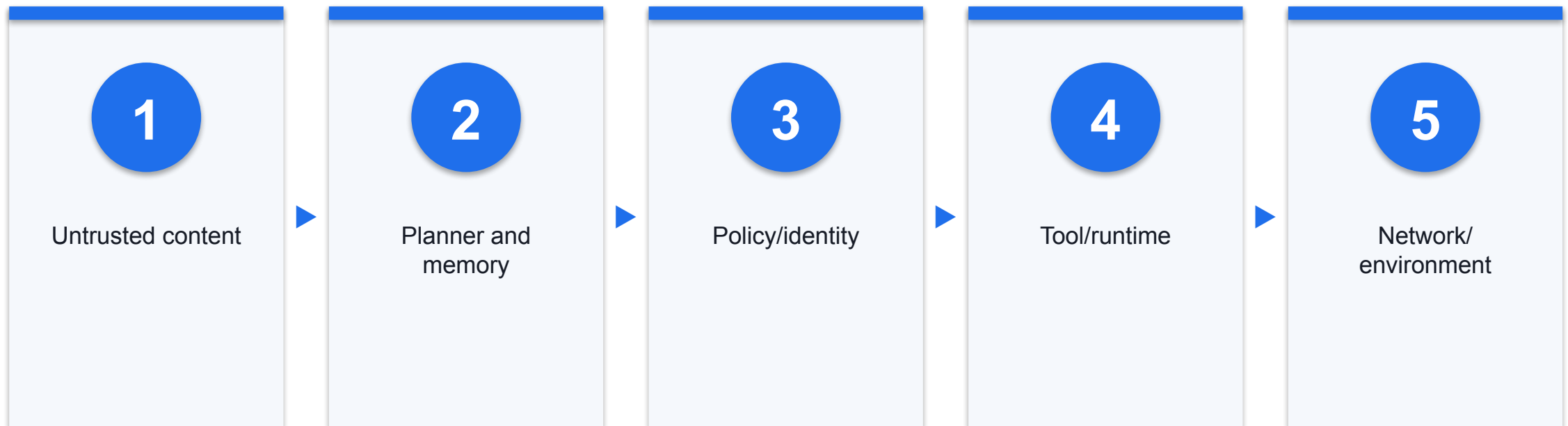
Outputs, actions, provenance and audit evidence

Trust Boundaries in a GenAI Application



Annotate who controls each boundary and which data crosses it.

Trust Boundaries in an AI Agent



The agent adds persistent state and action paths to the model boundary.

Treat Every Content Source as Untrusted

1

Human content

Prompts, email, chat and documents

2

Machine content

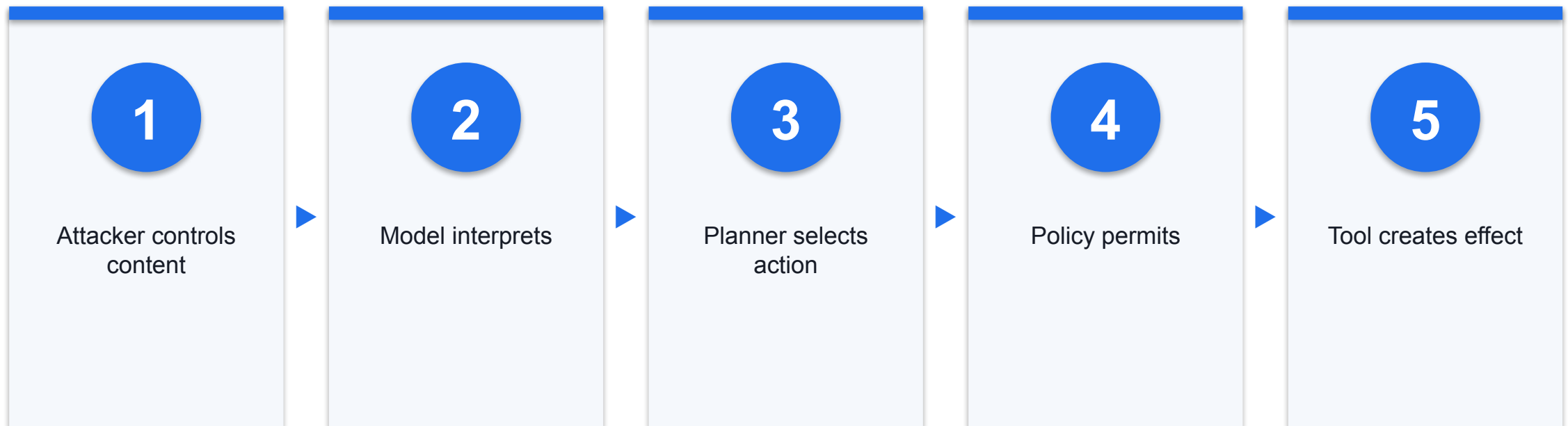
Web, APIs and tool output

3

Project content

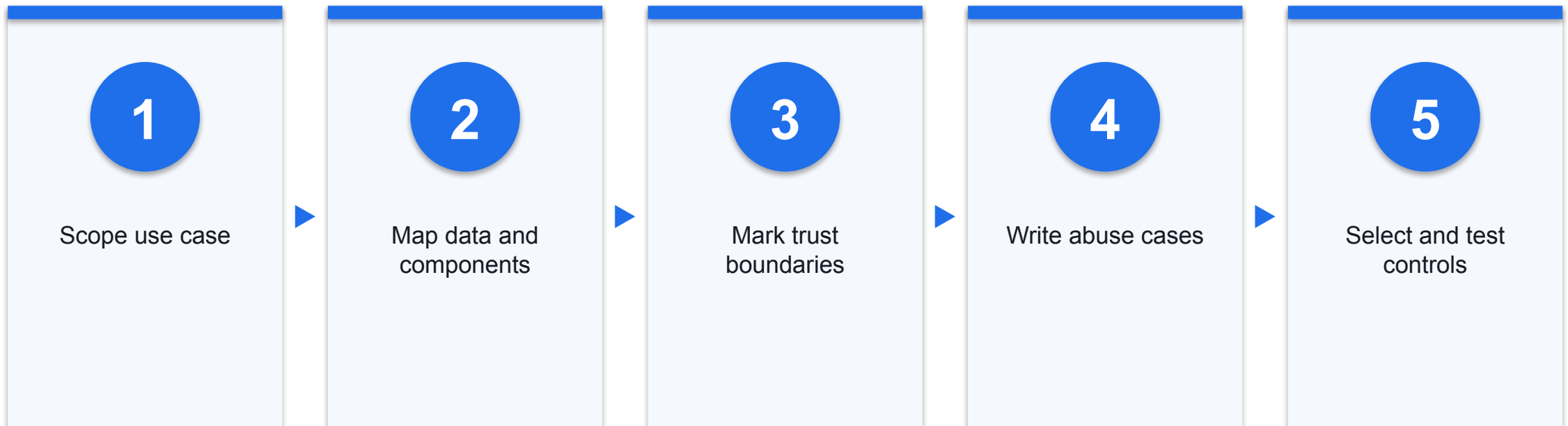
Memory, context files, skills, plugins and MCP metadata

Follow the Authority Path



Break the path with deterministic controls as close to the effect as possible.

Threat-Modelling Method

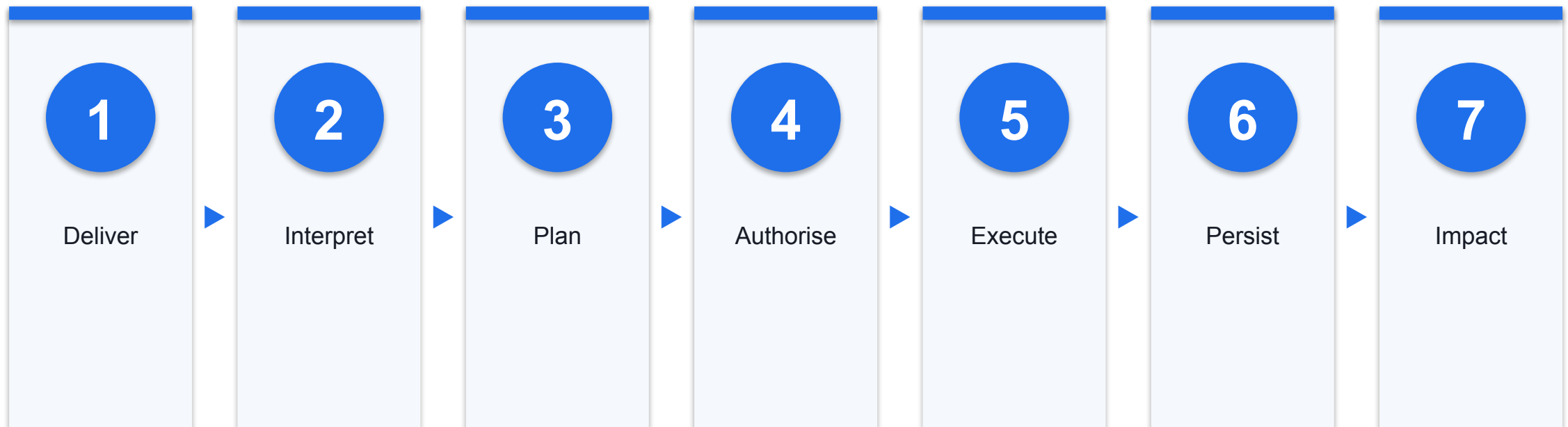


Attack Surface by Operating Layer

Surface	Generative	Agentic	Agent
Input	Prompt/context	Goal and observations	Channels, files, web
State	Context window	Loop state	RAG and persistent memory
Effect	Output	Plan/tool selection	Real system action

Evidence: SYN · Sources: S09, S11, S12

A Complete Attack Chain



Record evidence and a feasible control at every stage.

Activity 1 — Concierge Threat Model

Realistic synthetic classroom simulation

Map three input channels, four retrieval sources and four tools · Mark attacker-writable content and personal-data paths · Recommend proceed, conditional proceed or halt

1

Scope



2

Map



3

Test



4

Control



5

Evidence

DISCUSSION QUESTIONS

- Map three input channels, four retrieval sources and four tools
- Mark attacker-writable content and personal-data paths
- Recommend proceed, conditional proceed or halt

YOU'LL PRODUCE

A documented risk decision with evidence.

Assesses K2, K3, A4 · Activity pack: activities/activity-1-threat-modelling-genai-concierge/ · Full step-by-step facilitation detail: Learner Guide

Evidence: SIM

Activity 1 — Build the Data-Flow Diagram

Realistic synthetic classroom simulation

Place channels and data owners on the left · Put model, retrieval and memory in the centre · Put tool actions and affected people on the right

1

Scope



2

Map



3

Test



4

Control



5

Evidence

DISCUSSION QUESTIONS

- Place channels and data owners on the left
- Put model, retrieval and memory in the centre
- Put tool actions and affected people on the right

YOU'LL PRODUCE

A documented risk decision with evidence.

Assesses K2, K3, A4 · Activity pack: activities/activity-1-threat-modelling-genai-concierge/ · Full step-by-step facilitation detail: Learner Guide

Evidence: SIM

Activity 1 — Rank Trust Boundaries

Realistic synthetic classroom simulation

Who can write the source? · What authority can follow from it? · How reversible is the effect?

1

Scope



2

Map



3

Test



4

Control



5

Evidence

DISCUSSION QUESTIONS

- Who can write the source?
- What authority can follow from it?
- How reversible is the effect?

YOU'LL PRODUCE

A documented risk decision with evidence.

Assesses K2, K3, A4 · Activity pack: activities/activity-1-threat-modelling-genai-concierge/ · Full step-by-step facilitation detail: Learner Guide

Evidence: SIM

Activity 1 — Design Abuse Cases

Realistic synthetic classroom simulation

One GenAI content failure · One agentic loop or goal failure · One tool/identity action failure

1

Scope



2

Map



3

Test



4

Control



5

Evidence

DISCUSSION QUESTIONS

- One GenAI content failure
- One agentic loop or goal failure
- One tool/identity action failure

YOU'LL PRODUCE

A documented risk decision with evidence.

Assesses K2, K3, A4 · Activity pack: activities/activity-1-threat-modelling-genai-concierge/ · Full step-by-step facilitation detail: Learner Guide

Evidence: SIM

Activity 1 — Decide the Controls

Realistic synthetic classroom simulation

Rank controls by earliest break point · State latency, cost and user-friction trade-offs · Name evidence proving each control works

1

Scope

2

Map

3

Test

4

Control

5

Evidence

DISCUSSION QUESTIONS

- Rank controls by earliest break point
- State latency, cost and user-friction trade-offs
- Name evidence proving each control works

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A documented risk decision with evidence.

Assesses K2, K3, A4 · Activity pack: [activities/activity-1-threat-modelling-genai-concierge/](#) · Full step-by-step facilitation detail: [Learner Guide](#)

Evidence: SIM

Day 1 Integrated Risk Matrix

System	Primary exposure	Control direction
Generative	Content and disclosure	Constrain data access and downstream effect; validate before use
Agentic	Goal and loop	Stop conditions and budgets
Agent	Tools, identity, runtime	Least privilege, sandbox and approval

Evidence: SYN · Sources: S09, S11, S12, S14

Day 1 Recap — What Changed as AI Gained Agency?

1

Generate

Probability creates useful but fallible content

2

Iterate

Loops compound decisions and cost

3

Act

Tools and identity turn output into impact

Day 2 Preview — From Attack to Assurance



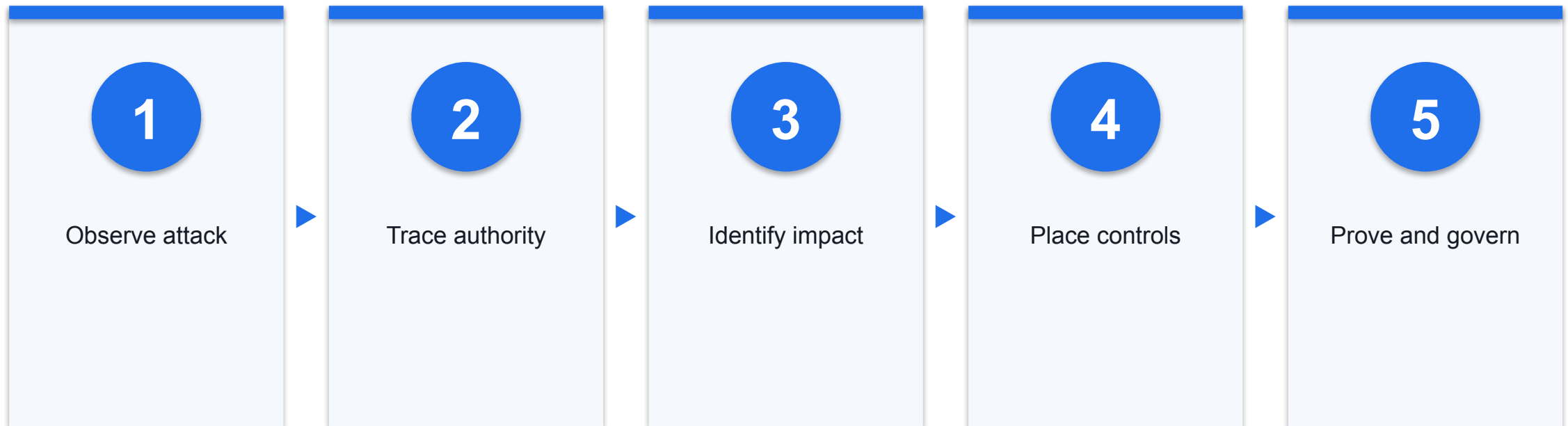
Evidence: ADMIN

COURSE MODULE

Day 2 — Attack, Defend, Govern, Operate

Cases make the mechanisms concrete; controls then become testable evidence.

Day 2 Dependency Map



Evidence: SYN · Sources: S10, S12, S14

COURSE MODULE

Prompt Injection and Jailbreaks

Distinguish changing application intent from bypassing model behaviour rules.

Prompt Engineering vs Prompt Injection

- Prompt engineering
- Authorised user shapes the requested output
- Success serves the declared task

- Prompt injection
- Untrusted text changes instructions or data flow
- Success serves an attacker or unintended objective

Direct Prompt Injection

1

Delivery

Attacker writes into the user-facing prompt

2

Conflict

Text asks the model to ignore or reinterpret policy

3

Control

Detect attempts and constrain downstream authority

Indirect Prompt Injection

1

Delivery

Payload sits in email, file, page or other content

2

Trigger

A legitimate user asks the system to process it

3

Impact

Model may treat data as instructions

Cross-Modal Prompt Injection

1

Carrier

Image, audio or mixed-media content

2

Recovery

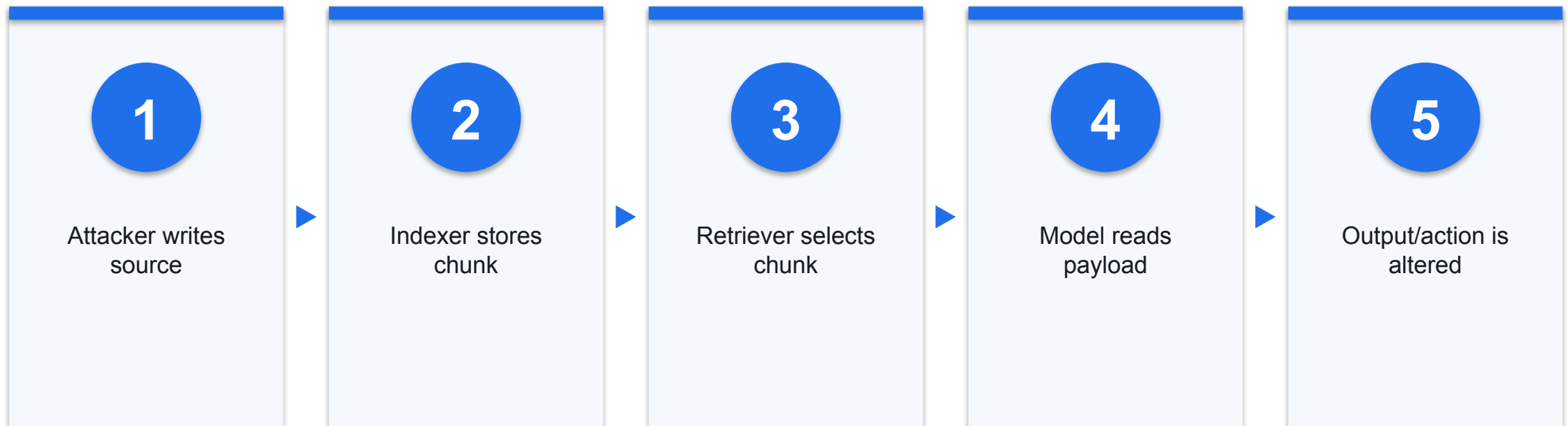
OCR, vision or transcription exposes instruction text

3

Gap

A text-only filter may never inspect the carrier

Retrieval-Augmented Injection



Secure both who may write the corpus and what authority follows retrieval.

Email and Document Injection

1

Legitimate task

Summarise a supplier email or invoice

2

Hostile content

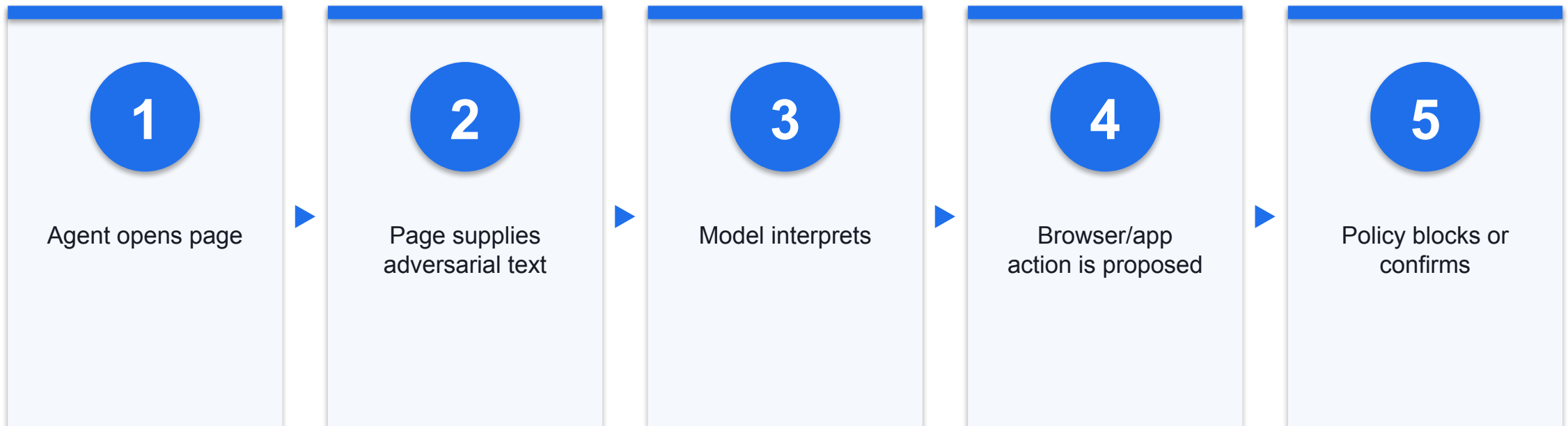
Quoted text requests data or a changed instruction

3

Boundary

Reader remains tool-free; sensitive actions use independent verification

Web and Browser Injection



Browsing expands the untrusted context continuously.

Context-File Injection

1

Discovery

Where configured, an agent discovers and loads project instruction files

2

Risk

A cloned repository may author instructions that influence tool use

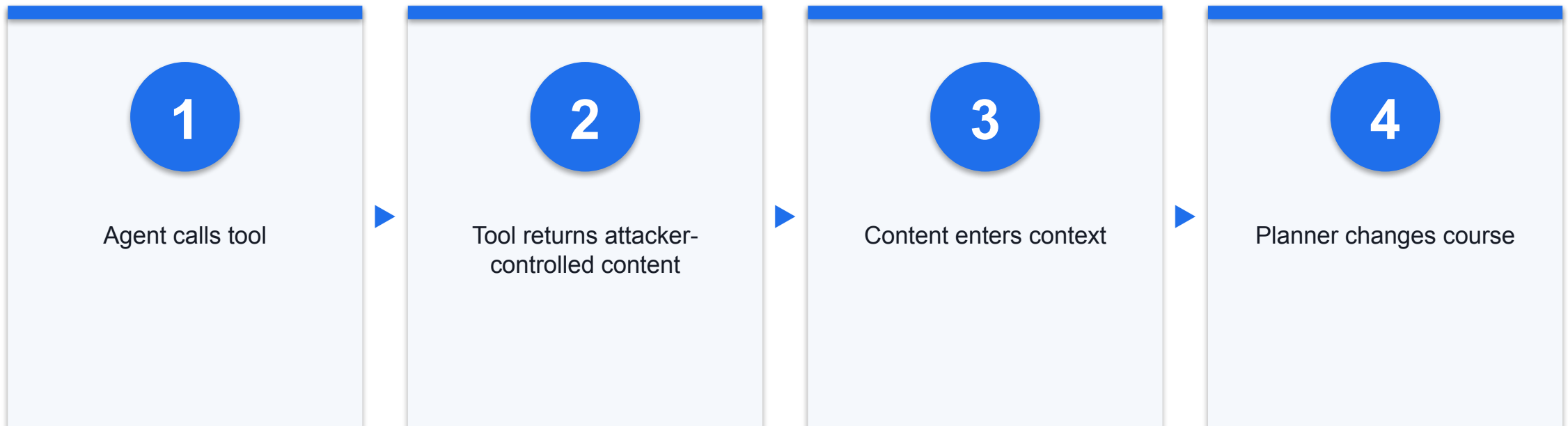
3

Control

Inspect, scope and sandbox untrusted projects before tool use

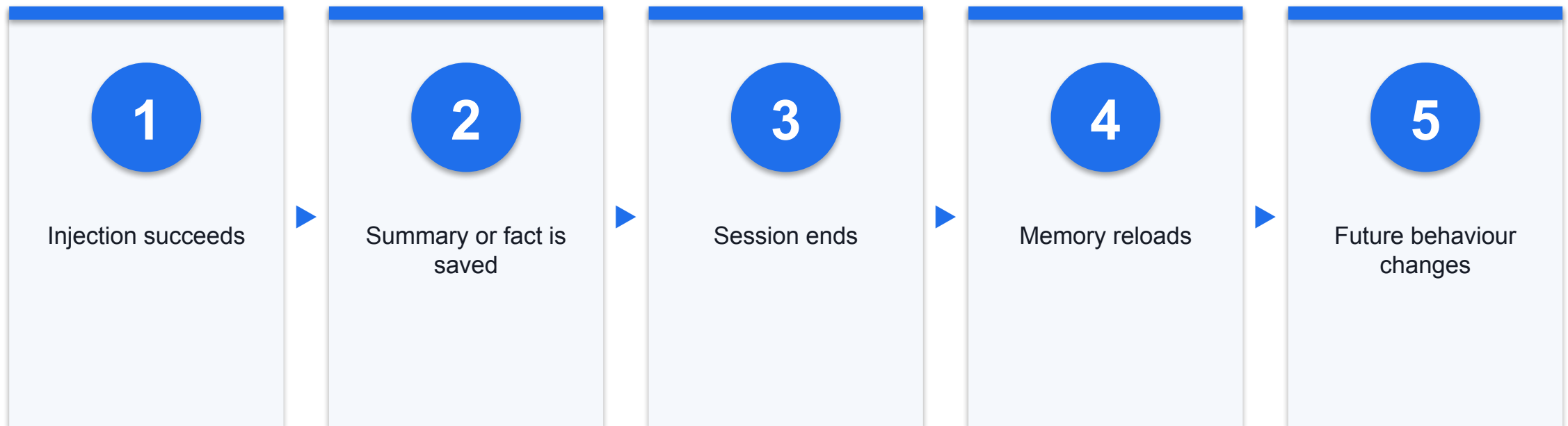
As of 20 Aug 2026; loading rules are product- and configuration-specific.

Tool-Output Injection



Tool output is data, not trusted policy.

Memory-Based Instruction Persistence



Memory write policy and provenance are security controls.

Jailbreak — Operational Definition

1

Objective

Bypass model or application behavioural restrictions

2

Method

Adversarial prompts, obfuscation or multi-turn pressure

3

Distinction

May not require changing a business task or calling a tool

Jailbreak vs Prompt Injection

Question	Jailbreak	Prompt injection
Target	Safety/behaviour restriction	Application instruction hierarchy
Typical carrier	User prompt	User or external content
Primary risk	Disallowed output	Changed task, disclosure or action

Evidence: DEF · Sources: S11, S13

Common Jailbreak Families — Course Synthesis

1

Framing

Roleplay or fictional authority

2

Obfuscation

Encoding, spacing or substitution

3

Composition

Split intent across turns or modalities

Why Model Guardrails Are Necessary but Insufficient

1

Useful

Block known patterns and create alert signals

2

Fallible

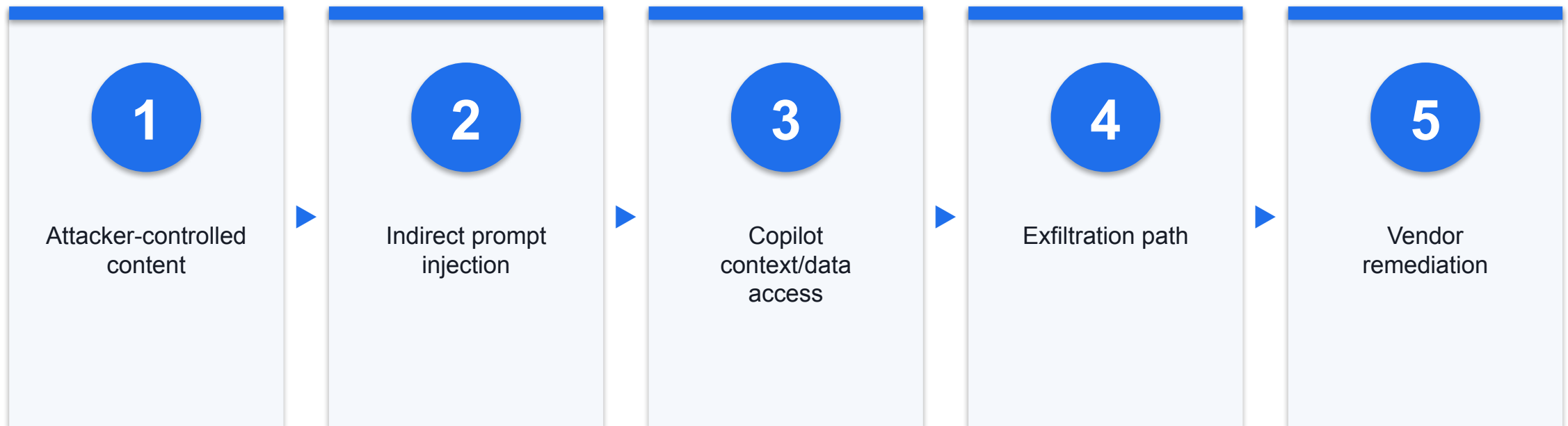
Novel phrasing and modalities can evade classifiers

3

Hard boundary

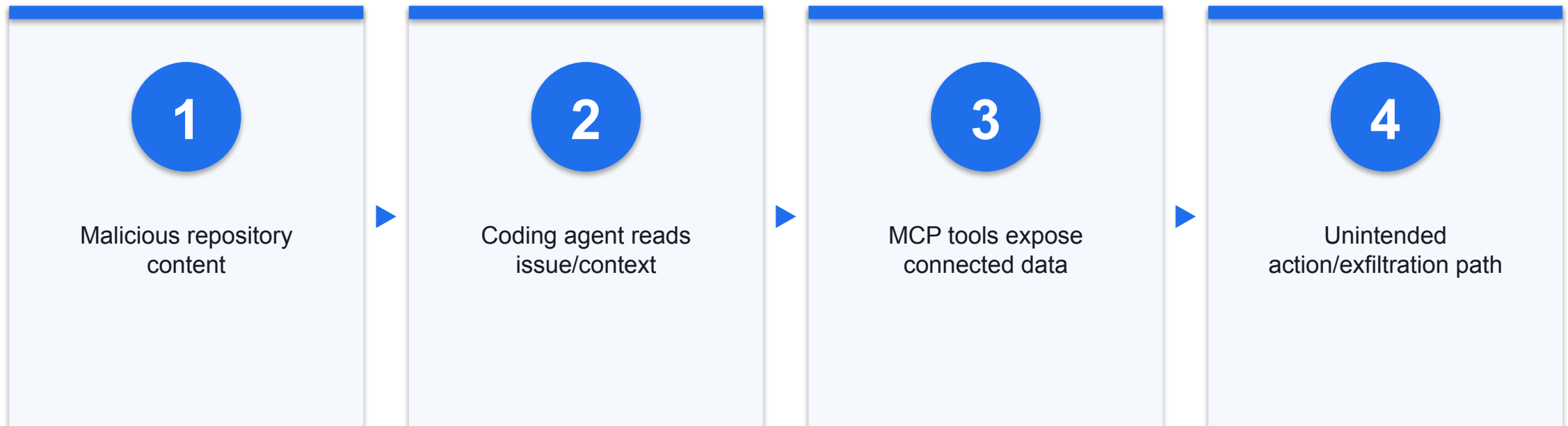
Policy, schema, least privilege and approval limit effects

Case — EchoLeak: Verified Vulnerability, Not a Confirmed Breach



Aim/Microsoft sources accessed 20 Aug 2026; verified vulnerability, not confirmed exploitation or a breach.

Published GitHub MCP Prompt-Injection Demonstration



Researcher demonstration; source accessed 20 Aug 2026. Production exploitation is not established.

Simulation — Employee Data Entered into Public GenAI

1

Action

Employee pastes confidential data into an external service

2

Risk

Disclosure, retention and purpose may exceed policy

3

Control

Approved service, minimisation, DLP and user training

Governance Case — Moffatt v Air Canada



Tribunal decision 2024 BCCRT 149; accessed 20 Aug 2026. Reliability/accountability case, not prompt injection or cyberattack.

Incident, Demonstration, Evaluation, or Hypothetical?

Label	Evidence threshold	How to speak
Incident	Observed production event	State confirmed scope
Demonstration	Reproduced attack path	State tested environment
Evaluation	Controlled model/system test	Do not call it a breach
Simulation	Fictional teaching case	Label every number synthetic

Evidence: SYN · Sources: S13

Activity 2 — Prompt-Injection Test

Realistic synthetic classroom simulation

Trace an indirect payload through a synthetic claims workflow · Measure task success and attack success separately · Select the earliest deterministic control that breaks the chain

1

Scope

2

Map

3

Test

4

Control

5

Evidence

DISCUSSION QUESTIONS

- Trace an indirect payload through a synthetic claims workflow
- Measure task success and attack success separately
- Select the earliest deterministic control that breaks the chain

YOU'LL PRODUCE

A documented risk decision with evidence.

Assesses K1, K4 · Activity pack: [activities/activity-2-prompt-injection-data-leakage/](#) · Full step-by-step facilitation detail: [Learner Guide](#)

Evidence: SIM

Activity 2 — Safe Harness and Test Boundaries

Realistic synthetic classroom simulation

Use only invented records · Write to a local dummy log · Block live email, shell, payment and production database tools

1

Scope

2

Map

3

Test

4

Control

5

Evidence

DISCUSSION QUESTIONS

- Use only invented records
- Write to a local dummy log
- Block live email, shell, payment and production database tools

YOU'LL PRODUCE

A documented risk decision with evidence.

Assesses K1, K4 · Activity pack: activities/activity-2-prompt-injection-data-leakage/ · Full step-by-step facilitation detail: Learner Guide

Evidence: SIM

Activity 2 — Establish a Clean Baseline

Realistic synthetic classroom simulation

Run legitimate tasks without hostile content · Record expected output and permitted action · Measure false blocks before adversarial changes

1

Scope

2

Map

3

Test

4

Control

5

Evidence

DISCUSSION QUESTIONS

- Run legitimate tasks without hostile content
- Record expected output and permitted action
- Measure false blocks before adversarial changes

YOU'LL PRODUCE

A documented risk decision with evidence.

Assesses K1, K4 · Activity pack: activities/activity-2-prompt-injection-data-leakage/ · Full step-by-step facilitation detail: Learner Guide

Evidence: SIM

Activity 2 — Test Controlled Variants

Realistic synthetic classroom simulation

Direct wording · Indirect document or email wording · Split and cross-modal variants

1

Scope

2

Map

3

Test

4

Control

5

Evidence

DISCUSSION QUESTIONS

- Direct wording
- Indirect document or email wording
- Split and cross-modal variants

YOU'LL PRODUCE

A documented risk decision with evidence.

Assesses K1, K4 · Activity pack: [activities/activity-2-prompt-injection-data-leakage/](#) · Full step-by-step facilitation detail: [Learner Guide](#)

Evidence: SIM

Activity 2 — Add Hard Controls

Realistic synthetic classroom simulation

Restrict retrieval to the current case · Validate recipients and tool parameters · Require approval for sensitive disclosure or write action

1

Scope

2

Map

3

Test

4

Control

5

Evidence

DISCUSSION QUESTIONS

- Restrict retrieval to the current case
- Validate recipients and tool parameters
- Require approval for sensitive disclosure or write action

YOU'LL PRODUCE

A documented risk decision with evidence.

Assesses K1, K4 · Activity pack: activities/activity-2-prompt-injection-data-leakage/ · Full step-by-step facilitation detail: Learner Guide

Evidence: SIM

Activity 2 — Read ASR and False Positives Together

Metric	Definition	Decision use
Attack success rate	Successful attacks / attempts	Residual adversarial exposure
False-positive rate	Blocked benign / benign tests	Business cost of guardrail
Clean-task success	Correct benign / benign tests	Utility under controls

Evidence: SYN · Sources: S09, S12



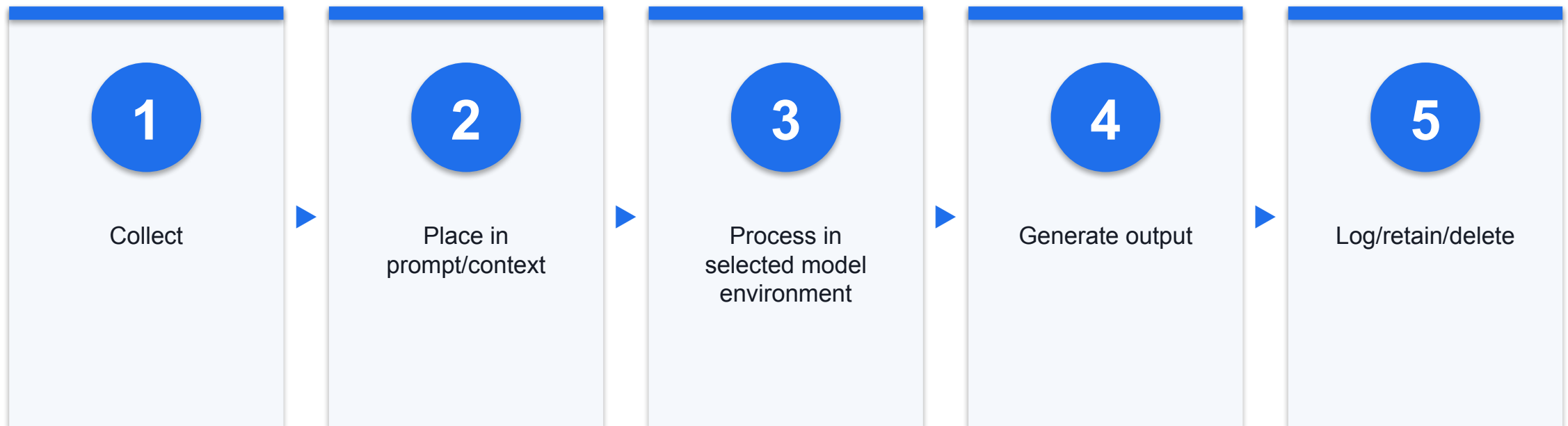
Morning Break

COURSE MODULE

Personal Data Across the AI Lifecycle

Map personal data wherever the system collects, derives, stores, transmits or logs it.

The GenAI Data Lifecycle



Purpose and protection duties apply across the lifecycle, not only at initial collection.

An Agent Can Add New Data Surfaces

1

Observation

Screenshots, browser state and external content

2

Action

Tool arguments, recipients and transaction details

3

Persistence

Memory, traces and scheduled-job state

Tokens Are Authorities, Not Just Strings

1

API key

Authorises service use and spend

2

OAuth token

Carries user-approved scopes to an app

3

Session credential

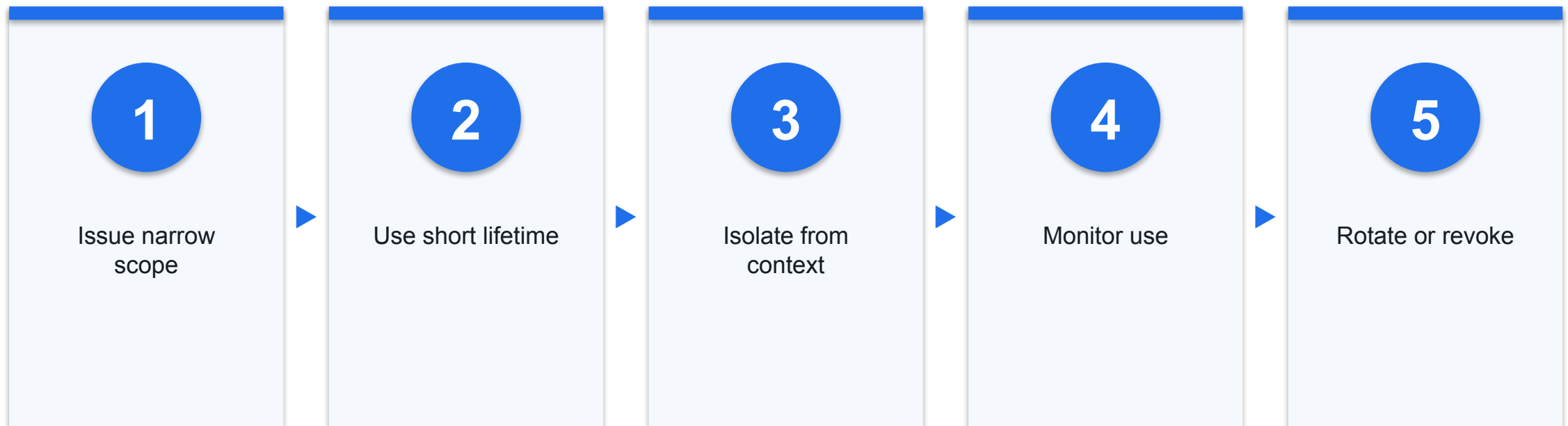
Represents an authenticated browser or gateway session

Token-Leak Paths

Path	Example	Control
Context/log	Secret pasted or returned by tool	Redact and block
Environment	Process inherits broad variables	Minimal environment
Browser	Cookie-bearing session	Dedicated profile and logout
Repository	Credential file in workspace	Secret scanning and isolation

Evidence: DEF · Sources: S17, S19, S20

Secret and Token Controls



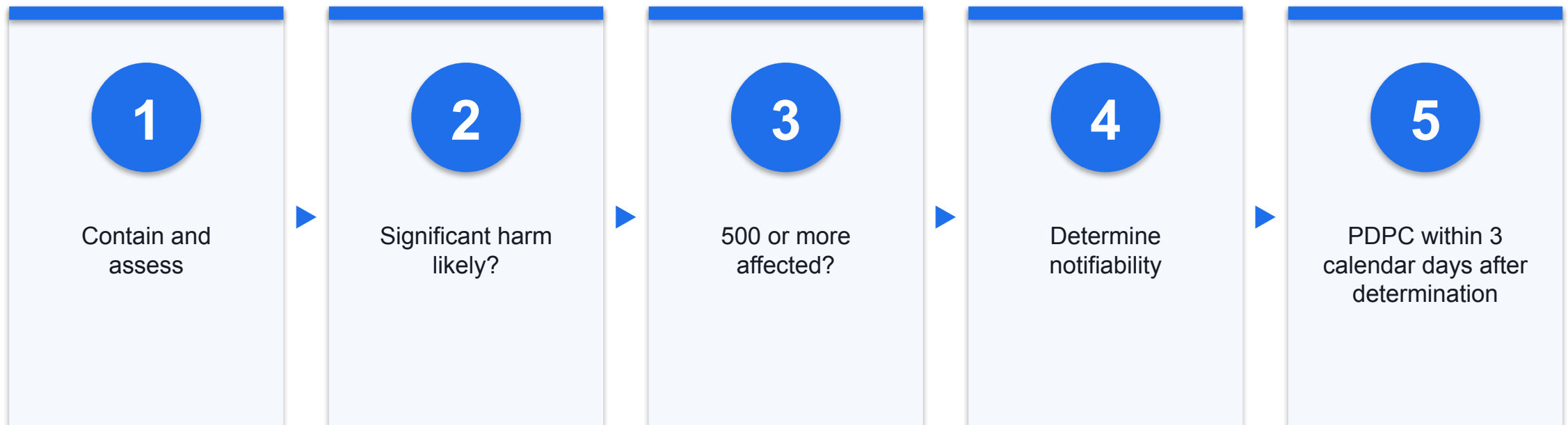
Never give the model authority that the containing process does not need.

PDPA Obligations Relevant to AI

Obligation	AI-system application
Notification / applicable basis	Document the legal basis or exception and notify purposes as required; consent is not universal
Purpose limitation	Use data only for appropriate notified purposes
Protection	Secure prompts, outputs, memory, tools and logs
Retention/transfer	Delete when no longer needed and protect overseas transfers
Accountability	Assign responsibility and policies

Evidence: DEF · Sources: S15

Data-Breach Notification Decision



Notify the PDPC if significant harm is likely OR the breach is of significant scale; notify affected individuals as soon as practicable where required.

Purpose Limitation and Data Minimisation

- Over-collection
- Whole mailbox or drive
- Full identifiers in prompts
- Indefinite memory

- Bounded design
- Task-specific folders/fields
- Tokenise or remove identifiers
- Purpose-linked retention

Vendors, Subprocessors, and Overseas Transfers

1

Data path

Where prompts, files, outputs and logs travel

2

Contract

Use, retention, deletion, subprocessors and incident support

3

Assurance

Comparable protection for overseas transfers

Logs, Retention, and Deletion

Need	Keep	Limit
Security	Identity, policy result, tool call and outcome	Raw personal content where unnecessary
Audit	Decision and approval evidence	Unlimited conversational history
Operations	Failure traces for defined period	Orphaned memories after decommission

Evidence: SYN · Sources: S15

Privacy by Design for Agents

1

Separate

Dedicated identities, stores and runtime boundaries

2

Minimise

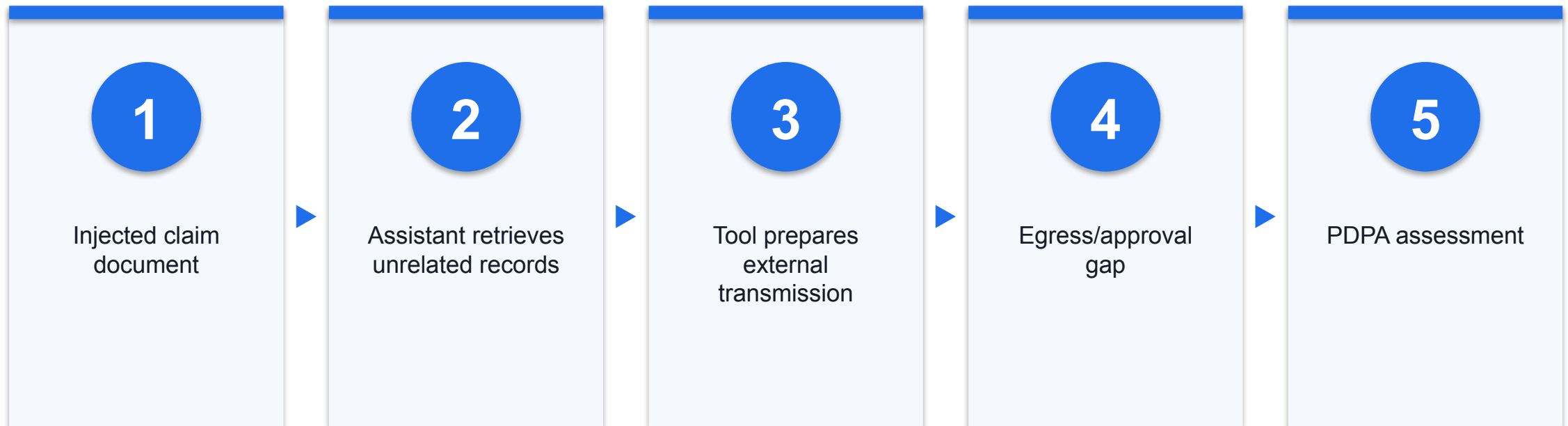
Only required fields enter model context

3

Control

Validate recipients and require review for sensitive disclosure

Fictional Case — Singapore Claims Assistant Leak



All names, counts, systems and outcomes are fictional.

Data-Leakage Mitigation Playbook



Evidence: SYN · Sources: S15, S16

Evidence Exercise — Build the Agent Data Inventory

- List each field and whether it is personal data
- Name collection source, purpose, recipient and storage
- Set access, retention, deletion and breach owner



Lunch Break

COURSE MODULE

Frameworks Before Incident Analysis

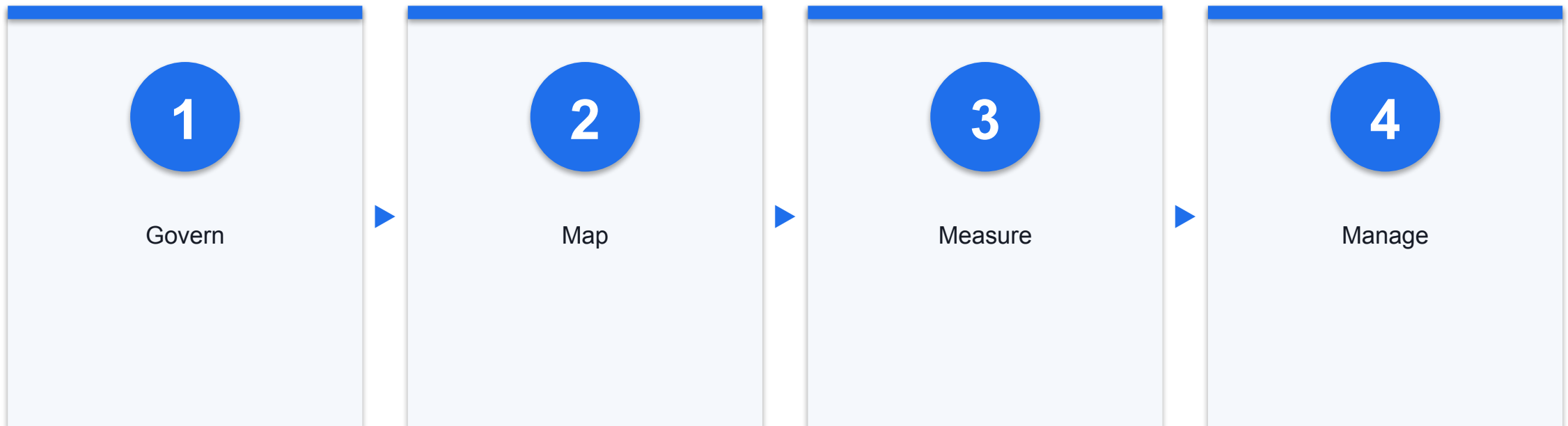
Use each framework for the question it was designed to answer.

The Framework Stack — Different Instruments

Instrument	Primary question	Use
NIST AI RMF	How do we manage risk?	Lifecycle
OWASP	What can go wrong?	Risk / vulnerability taxonomy
MITRE ATLAS	How do adversaries operate?	Techniques
IMDA	How do we bound and govern agents?	Accountability
PDPA	What duties apply to personal data?	Legal obligation

Evidence: DEF · Sources: S10, S11, S12, S13, S14, S15

NIST AI RMF — Govern, Map, Measure, Manage



The functions are iterative; governance is cross-cutting rather than a final sign-off.

OWASP LLM and Agentic Top 10

- LLM application
- Prompt, data, model and output risks
- Useful for GenAI/RAG applications

- Agentic application
- Goals, tools, identity, memory and coordination
- Useful when systems plan and act

MITRE ATLAS

1

Knowledge base

Tactics and techniques against AI-enabled systems

2

Coverage

Maps tactics, techniques, mitigations and documented case studies

3

Use

Threat assessment, red teaming and detection design

IMDA Agentic AI Governance Framework — Four Dimensions

1 Assess and bound

Choose suitable use cases; limit action space, data, tools and autonomy

2 Human accountability

Define significant checkpoints and meaningful approval

3 Technical controls

Apply controls and processes across the lifecycle

4 End-user responsibility

Enable informed use through transparency and training

Action space and autonomy are distinct; some use cases may be unsuitable for agents.

PDPA and PDPC Guidance

Duty	Security implication
Purpose/notification	Document why data enters the system
Protection	Secure every prompt, memory, output and tool path
Retention/transfer	Control duration and overseas handling
Breach notification	Assess and notify when statutory tests are met

Use enacted PDPA obligations and published PDPC guidance. The 2 Jun 2026 GenAI document is a public-consultation proposal, not final guidance.

Positive Governance Case — Dayos Tiered IT Actions

Tier	Documented pattern	Control lesson
Tier 1	Low severity and reversible; propose-confirm with requesting user; no human engineer	Record reasoning/confidence; audit a cross-section biweekly
Tier 2	Agent diagnoses and proposes; human approval is required	Place approval before the effect
Tier 3	High severity and limited reversibility	Agent cannot perform the action

IMDA-contributed case study, not an independent security endorsement; accessed 20 Aug 2026.

OpenClaw Threat Walkthrough

Boundary	Documented fact	Security decision
Trust	One trusted operator per gateway	Separate mixed-trust users
Tools	High-impact control-plane/node tools exist	Deny by default for untrusted content
State	Runtime directory contains sensitive material	Protect host and file permissions

As of 20 Aug 2026; this is a configuration walkthrough, not a universal risk rating.

Hermes Threat Walkthrough

Boundary	Documented fact	Security decision
Execution	Local and isolated backends are available	Choose backend by risk
Context	Project files and skills can shape behaviour	Scan and review
Tools	Approvals and filters provide layers	Keep high-risk tools disabled unless needed

As of 20 Aug 2026; selected backend, approvals and tool configuration determine exposure.

COURSE MODULE

When AI Can Act: Agentic Threats

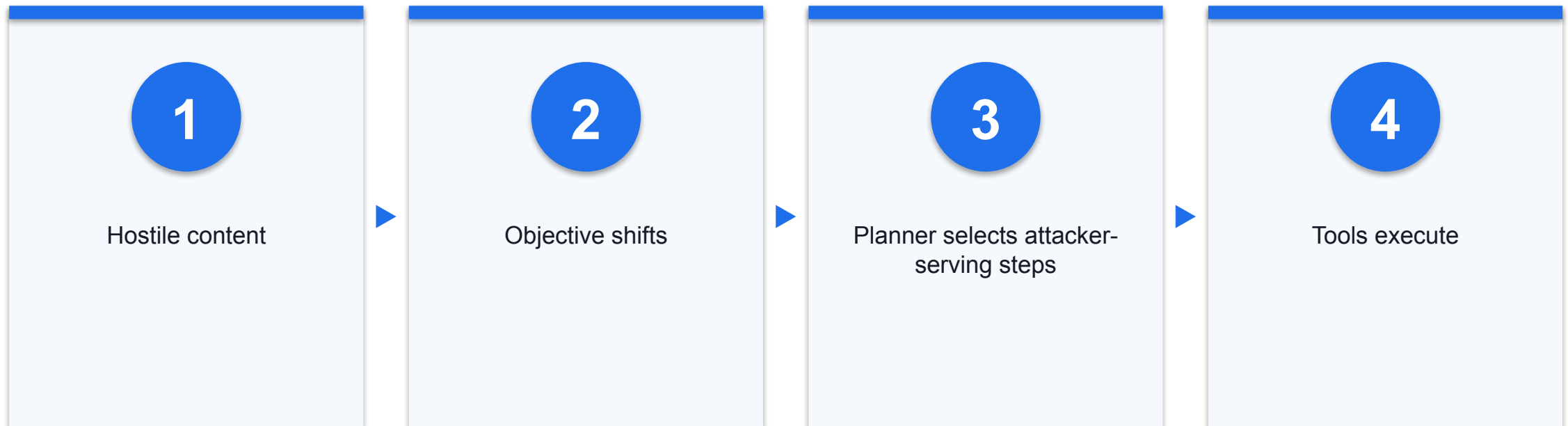
Map each threat to a component, authority path and earliest control point.

OWASP Agentic Top 10 — System View

Component	Representative risks
Goal/planner	Goal hijack, rogue behaviour
Tools/identity	Tool misuse, privilege abuse, code execution
State/comms	Memory poisoning, insecure inter-agent communication
System/human	Supply chain, cascades, trust exploitation

Evidence: DEF · Sources: S12

ASI01 — Agent Goal Hijack



Bind goals and validate actions independently of the model's stated rationale.

ASI02 — Tool Misuse

1

Legitimate tool

Tool itself may work as designed

2

Harmful use

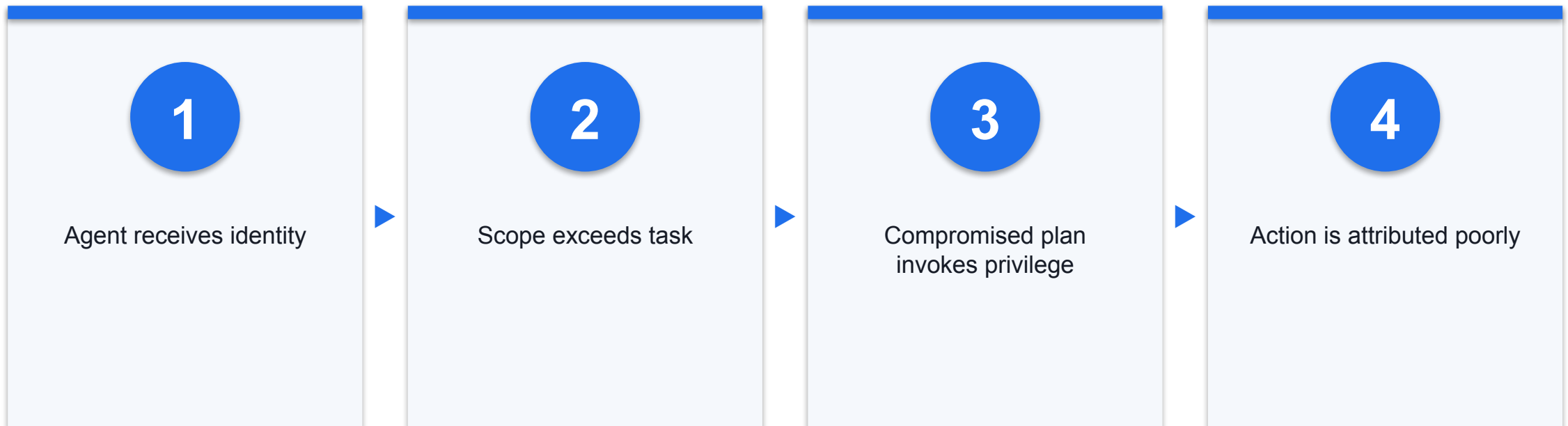
Parameters, sequence or context creates harm

3

Control

Allowlist, schema, policy and approval

ASI03 — Identity and Privilege Abuse



Use a distinct agent identity with task-specific, short-lived permissions.

ASI04 — Agentic Supply-Chain Vulnerabilities

1

Components

Models, skills, plugins, MCP, packages and prompts

2

Change

Runtime discovery and updates can alter reviewed behaviour

3

Control

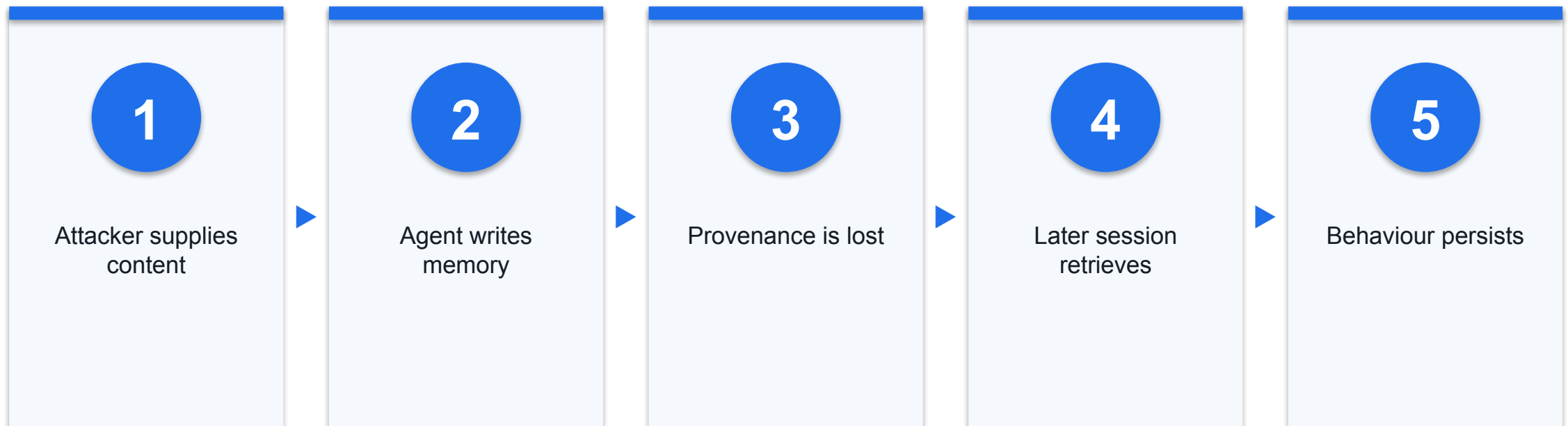
Provenance, pinning, integrity, isolation and inventory

ASI05 — Unexpected Code Execution

Path	Exposure	Boundary
Generated code	Unsafe program is proposed	Review and test
Interpreter/shell	Code is executed	Sandbox and allowlist
Host escape	Runtime boundary fails	Hardened isolation and monitoring

Evidence: DEF · Sources: S12

ASI06 — Memory and Context Poisoning



Gate memory writes and retain source, author, timestamp and review status.

ASI07 — Insecure Inter-Agent Communication

1

Identity

Can the receiver authenticate the sender?

2

Message

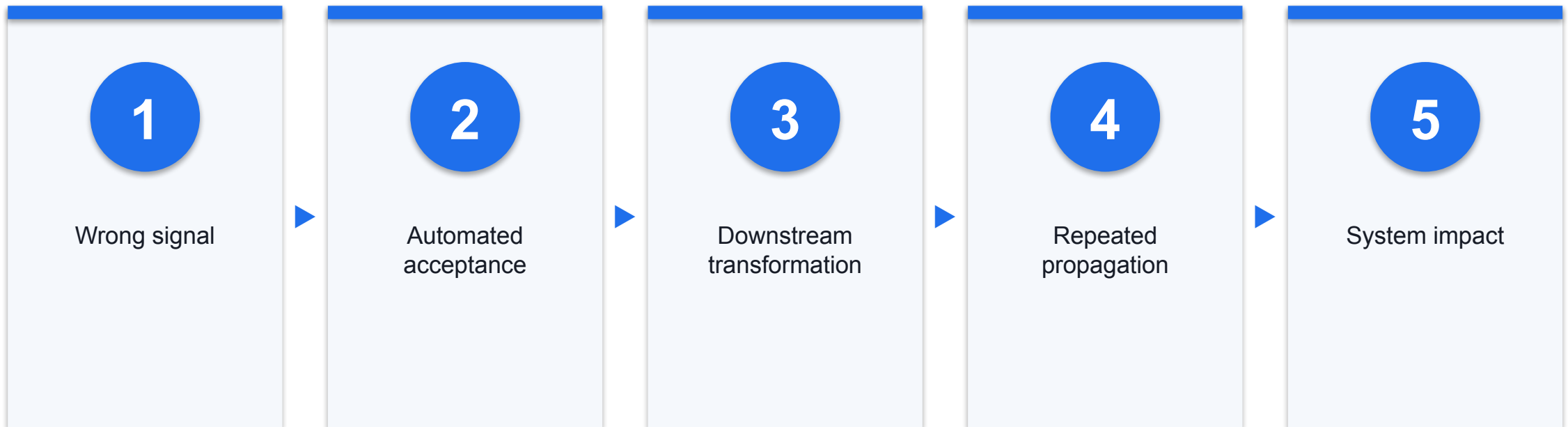
Is task intent and data integrity protected?

3

Delegation

Does authority narrow across handoffs?

ASI08 — Cascading Failures



Validate at handoff boundaries and set fan-out, retry and budget limits.

ASI09 — Human-Agent Trust Exploitation

1

Presentation

Confident explanation appears authoritative

2

Human factor

Time pressure and automation bias weaken review

3

Control

Show evidence and exact effect; audit approval quality

ASI10 — Rogue Agents

- Supported statement
- OWASP identifies rogue-agent risk
- Evaluations may test concealment or misalignment

- Unsupported leap
- An evaluation is a production breach
- A model finding proves every deployment behaves alike

Malicious or Untrusted Skills

1

Instructions

A skill can steer tool selection and behaviour

2

Code

Bundled scripts can execute with host authority

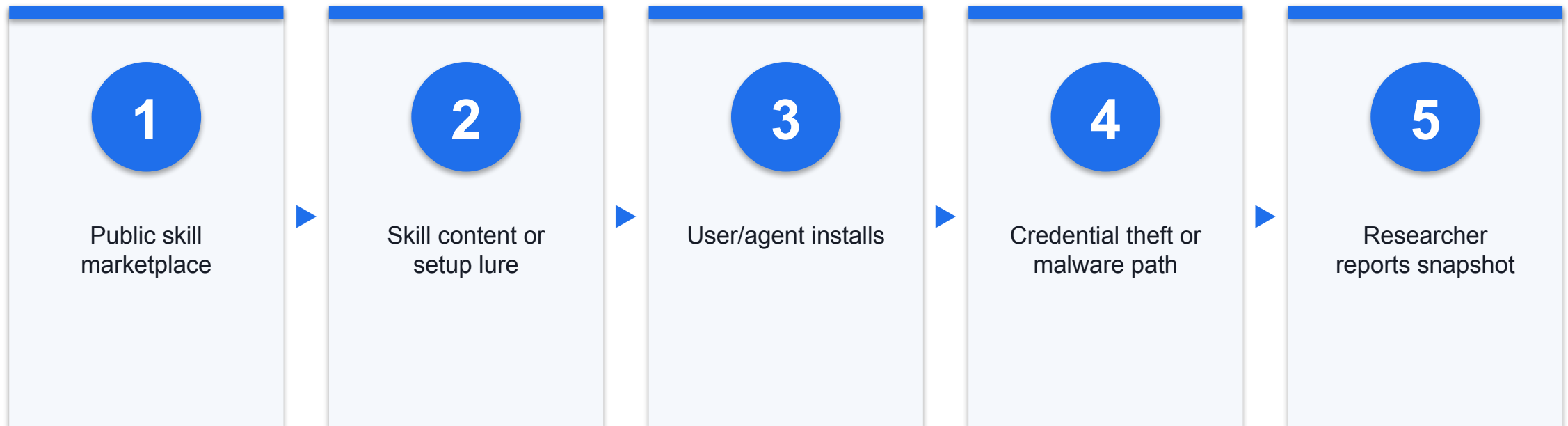
3

Control

Review source, content, version, capabilities and updates

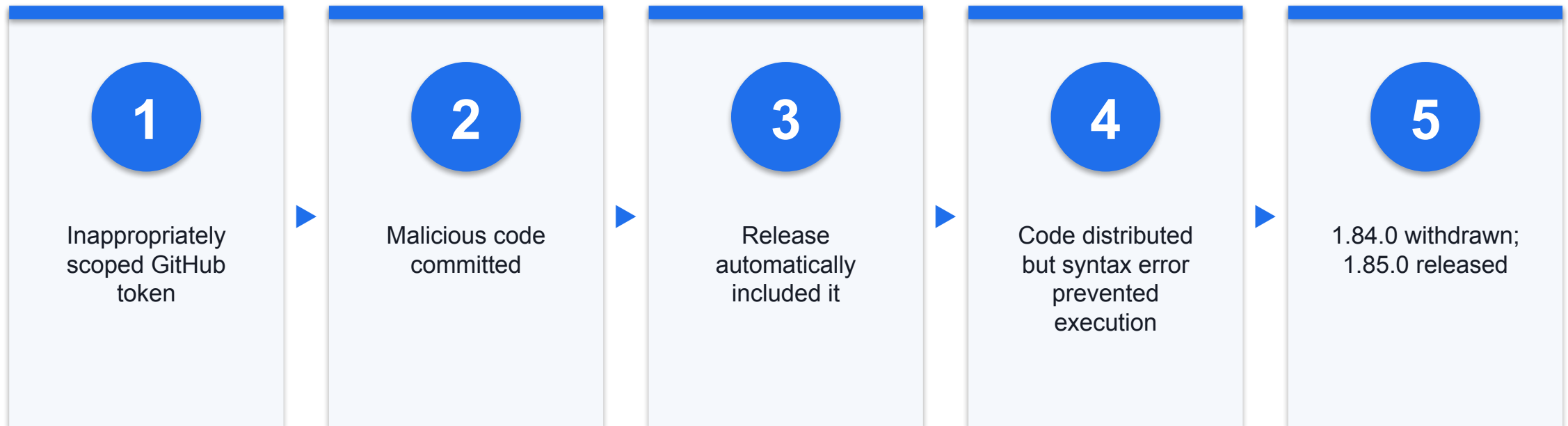
Evidence: DEF · Sources: S12, S18, S19

CASE-R — ClawHavoc Reported Malicious-Skill Campaign



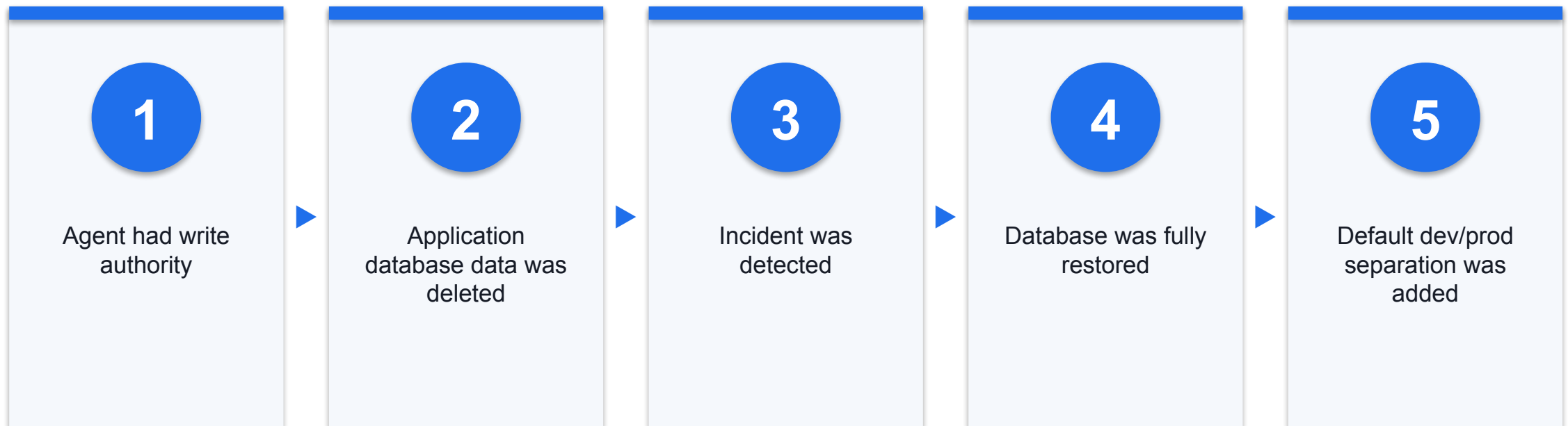
Koi, 1 Feb 2026; accessed 20 Aug 2026: 341 of 2,857 skills in its audited snapshot, including 335 linked to one campaign; not a current ecosystem rate or affected-user count.

CASE-V — Amazon Q Developer Extension 1.84.0



AWS bulletin published 23 Jul 2025, updated 25 Jul 2025; distribution is not proof of execution or customer compromise.

CASE-V — Replit Agent and an Application Database



Replit first-party post published 29 Jul 2025; it says no data was lost. Do not describe a permanent loss, customer breach or primary company database.

Network Exposure

Surface	Question	Control
Listener/gateway	Who can reach it?	Bind, authenticate, segment
Messaging channel	Who can trigger tools?	Pair and allowlist
Outbound web/API	Where can data go?	Proxy and destination allowlist

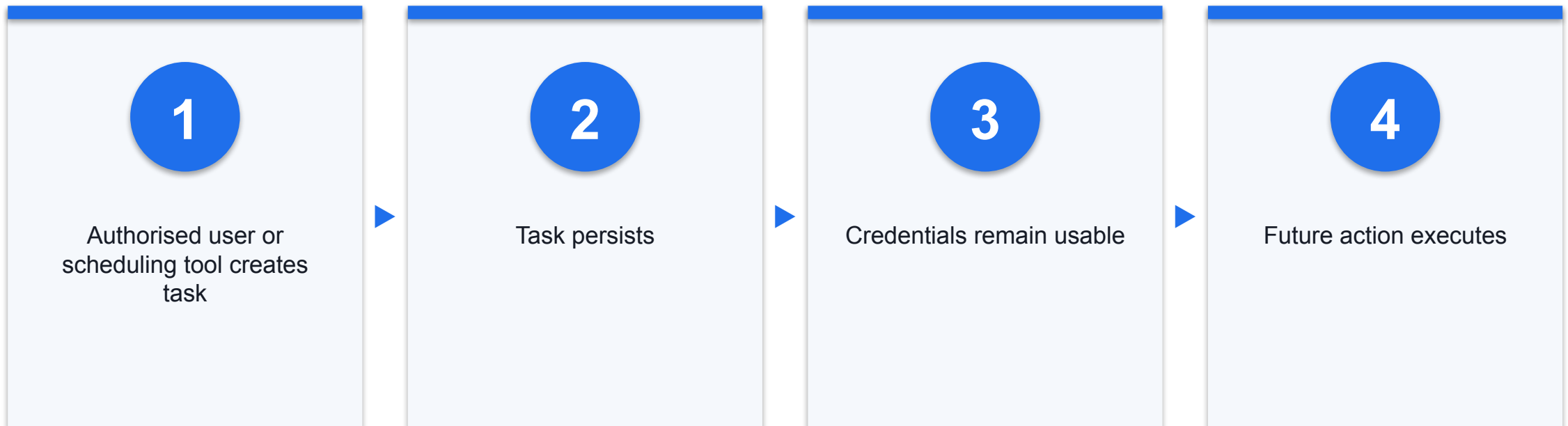
Evidence: DEF · Sources: S17, S19

Egress, Code Execution, and Host Impact

- Compromise without hard boundaries
- Broad network
- User-level shell
- Long-lived credentials

- Contained design
- Destination allowlist
- Non-root sandbox
- Short-lived scoped credentials

Where Scheduling Is Enabled

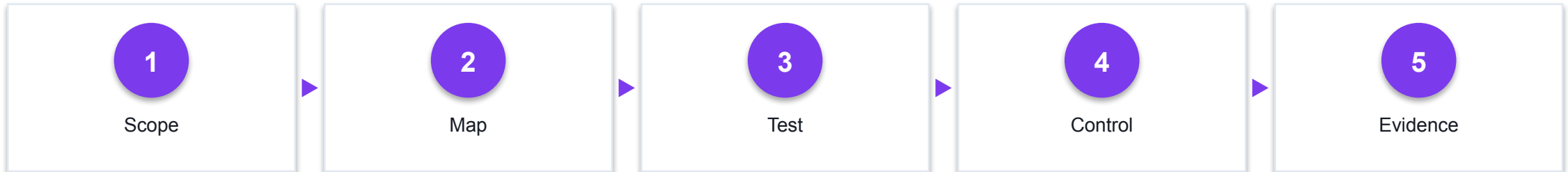


Inventory schedules, require approval, cap lifetime and provide revocation.

Activity 3 — Framework Stack

Realistic synthetic classroom simulation

Use a fictional clinic assistant plus billing agent · Match each decision question to a framework · Avoid claiming that any one framework is complete



DISCUSSION QUESTIONS

- Use a fictional clinic assistant plus billing agent
- Match each decision question to a framework
- Avoid claiming that any one framework is complete

YOU'LL PRODUCE

A documented risk decision with evidence.

Assesses A3, A5 · Activity pack: [activities/activity-3-security-framework-selection/](#) · Full step-by-step facilitation detail: [Learner Guide](#)

Evidence: SIM

Activity 3 — Threat and Metric Map

Realistic synthetic classroom simulation

Map an OWASP risk to an ATLAS technique only where direct evidence supports it; otherwise record no direct mapping · Map controls to NIST/IMDA lifecycle evidence · Attach PDPA duties where personal data is involved

1

Scope



2

Map



3

Test



4

Control



5

Evidence

DISCUSSION QUESTIONS

- Map an OWASP risk to an ATLAS technique only where direct evidence supports it; otherwise record no direct mapping
- Map controls to NIST/IMDA lifecycle evidence
- Attach PDPA duties where personal data is involved

YOU'LL PRODUCE

A documented risk decision with evidence.

Assesses A3, A5 · Activity pack: [activities/activity-3-security-framework-selection/](#) · Full step-by-step facilitation detail: [Learner Guide](#)

Evidence: SIM

Activity 3 — Go-Live Gate

Realistic synthetic classroom simulation

Define clean-task and adversarial tests · Name the accountable risk owner · Set conditions for go, conditional go or no-go

1

Scope

2

Map

3

Test

4

Control

5

Evidence

DISCUSSION QUESTIONS

- Define clean-task and adversarial tests
- Name the accountable risk owner
- Set conditions for go, conditional go or no-go

YOU'LL PRODUCE

A documented risk decision with evidence.

Assesses A3, A5 · Activity pack: [activities/activity-3-security-framework-selection/](#) · Full step-by-step facilitation detail: [Learner Guide](#)

Evidence: SIM

Activity 4 — Case Review

Evidence-based documented case review

EchoLeak: CASE-V verified vulnerability, not confirmed breach · Amazon Q extension and Replit database event: CASE-V first-party accounts · ClawHavoc: CASE-R research snapshot with denominator and limitations

1

Scope

2

Map

3

Test

4

Control

5

Evidence

DISCUSSION QUESTIONS

- EchoLeak: CASE-V verified vulnerability, not confirmed breach
- Amazon Q extension and Replit database event: CASE-V first-party accounts
- ClawHavoc: CASE-R research snapshot with denominator and limitations

YOU'LL PRODUCE

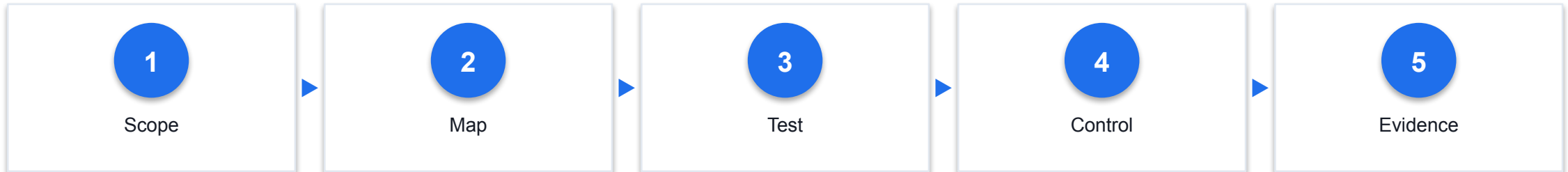
A documented risk decision with evidence.

Assesses K5 · Activity pack: [activities/activity-4-rogue-agent-incident-review/](#) · Full step-by-step facilitation detail: [Learner Guide](#)

Activity 4 — Kill Chain and Supply Gate

Evidence-based documented case review

Reconstruct delivery through impact · Find the earliest effective break point · Apply provenance, integrity, capability, isolation, approval and operations gates



DISCUSSION QUESTIONS

- Reconstruct delivery through impact
- Find the earliest effective break point
- Apply provenance, integrity, capability, isolation, approval and operations gates

YOU'LL PRODUCE

A documented risk decision with evidence.

Assesses K5 · Activity pack: [activities/activity-4-rogue-agent-incident-review/](#) · Full step-by-step facilitation detail: [Learner Guide](#)

Evidence: SIM



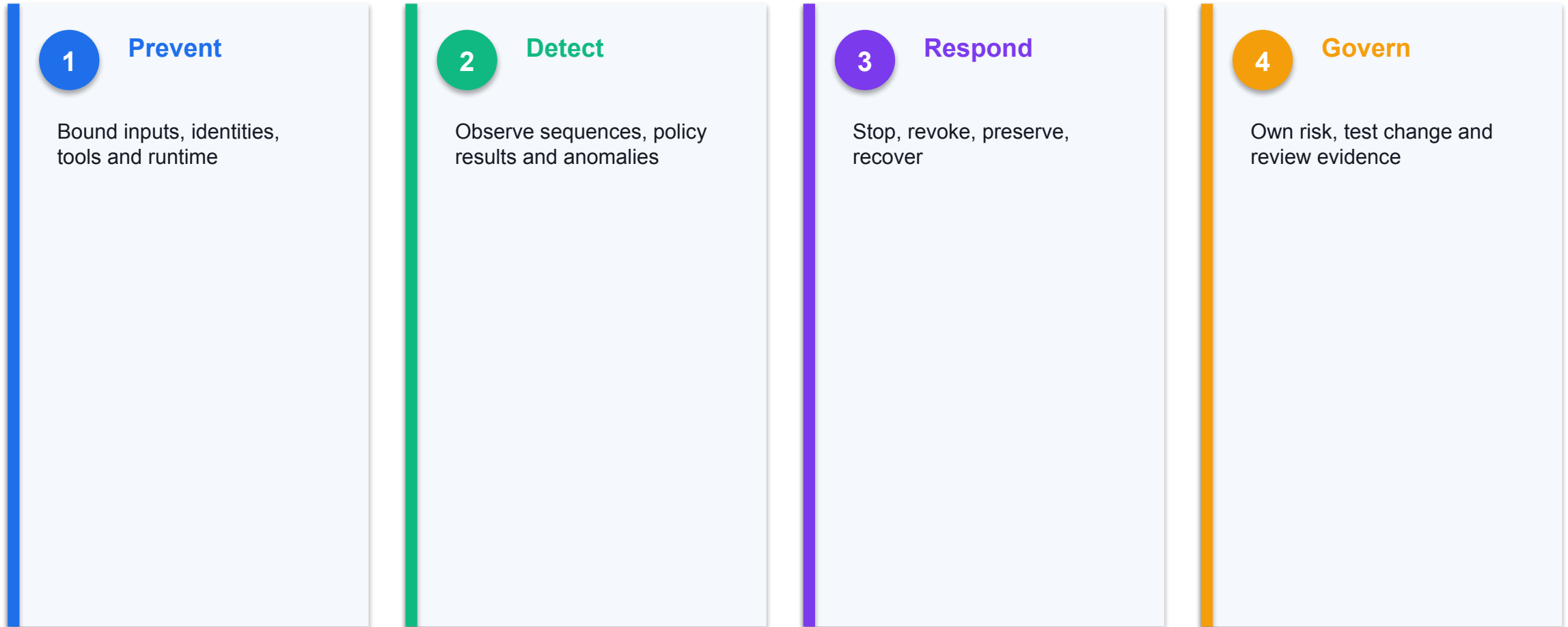
Afternoon Break

COURSE MODULE

From Threats to an Operating Control System

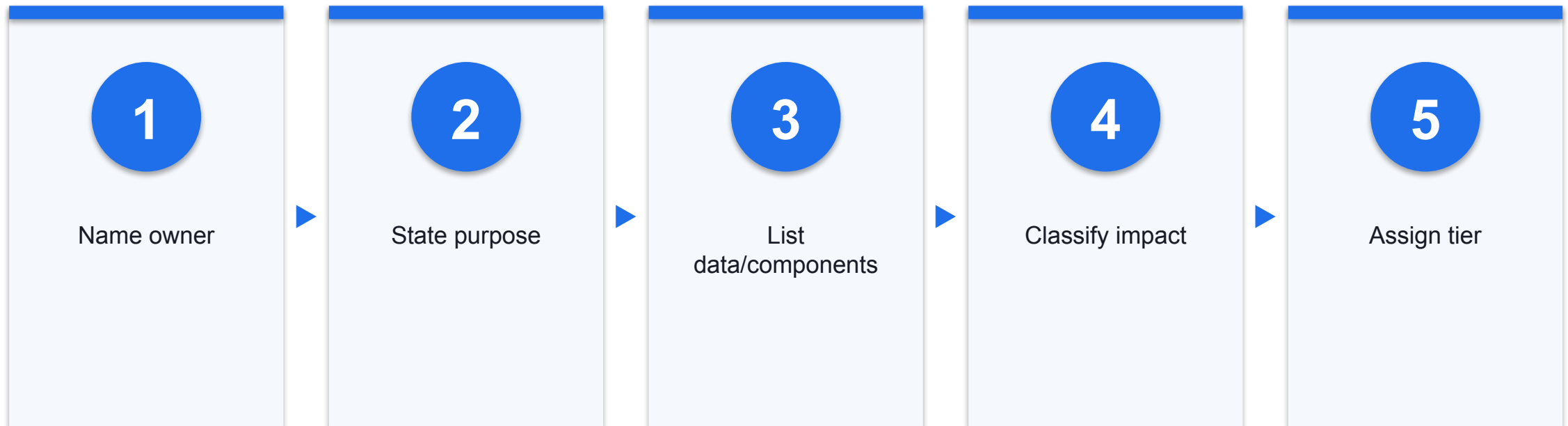
Controls must prevent, detect, respond and leave evidence.

Defence in Depth for GenAI and Agents



Evidence: SYN · Sources: S10, S12, S14

Step 1 — Inventory and Risk-Tier the Use Case



Step 2 — Bound Data, Tools, and Autonomy

1

Data

Allowed sources, fields and retention

2

Tools

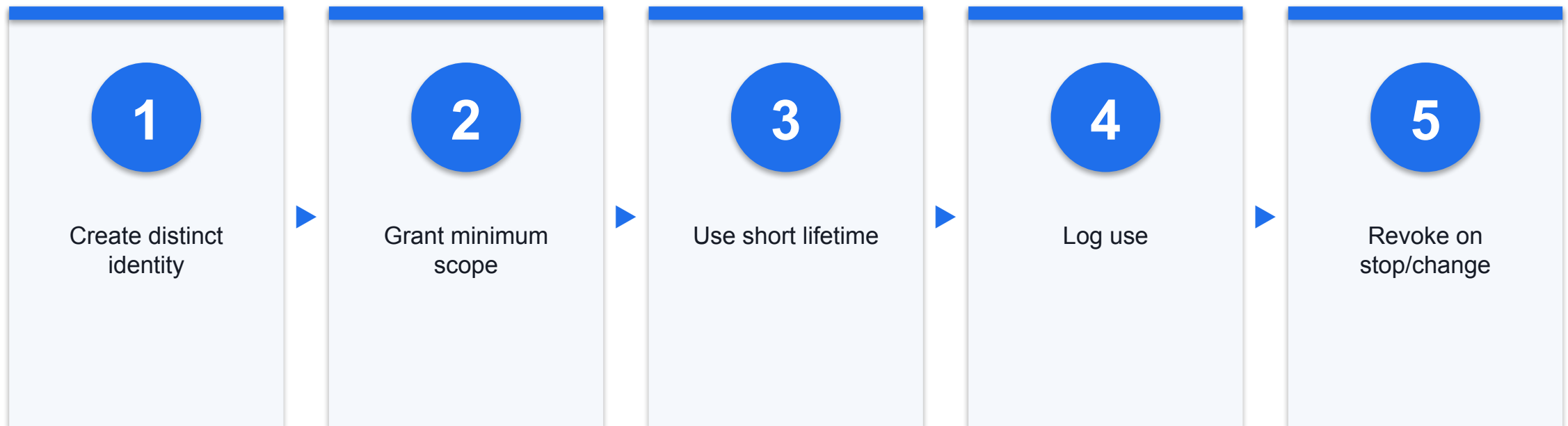
Allowed operations and parameter ranges

3

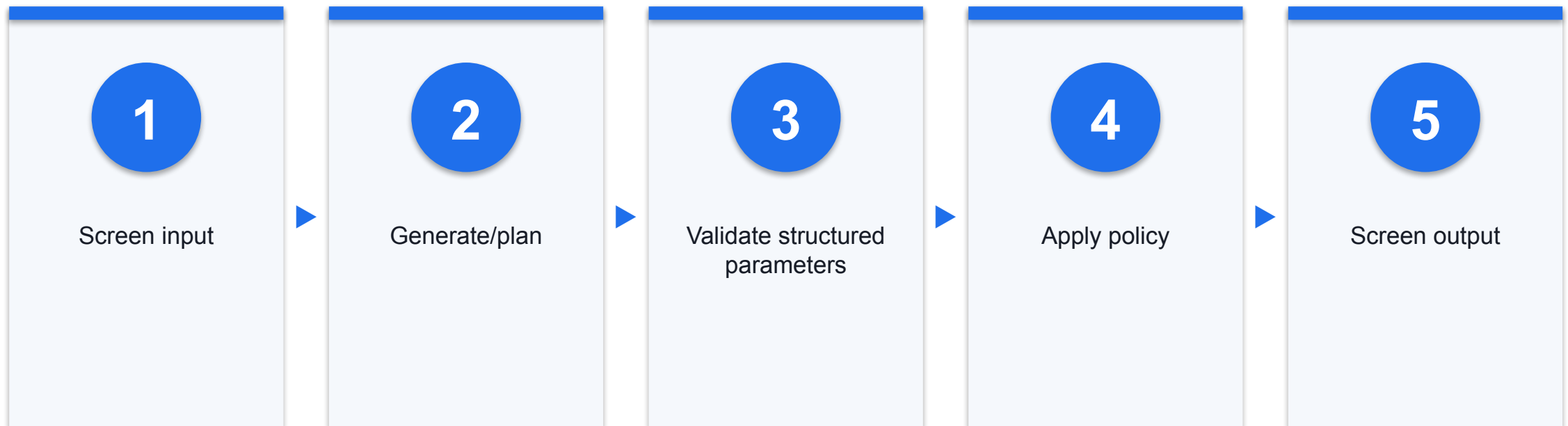
Autonomy

Stop conditions, budgets and approval points

Step 3 — Give Every Agent a Scoped Identity



Step 4 — Guardrails and Schema Validation



Model-based filters add friction; deterministic checks bound actions.

Step 5 — Sandbox Code and Files

1

Process

For containerised workloads, prefer non-root and no-new-privileges where supported

2

Filesystem

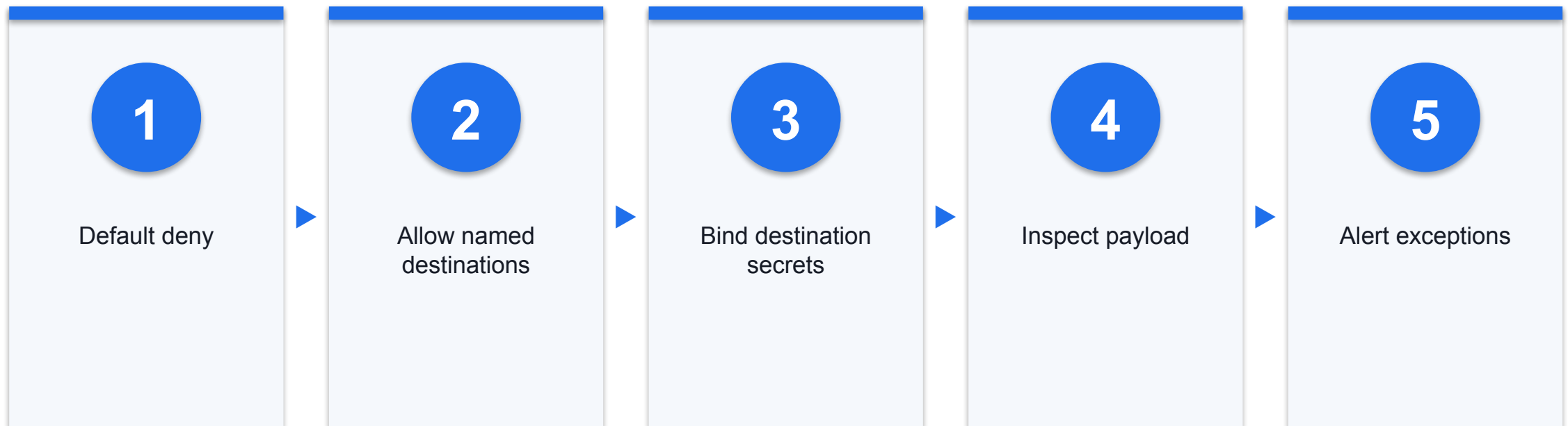
Workspace-only or read-only mounts

3

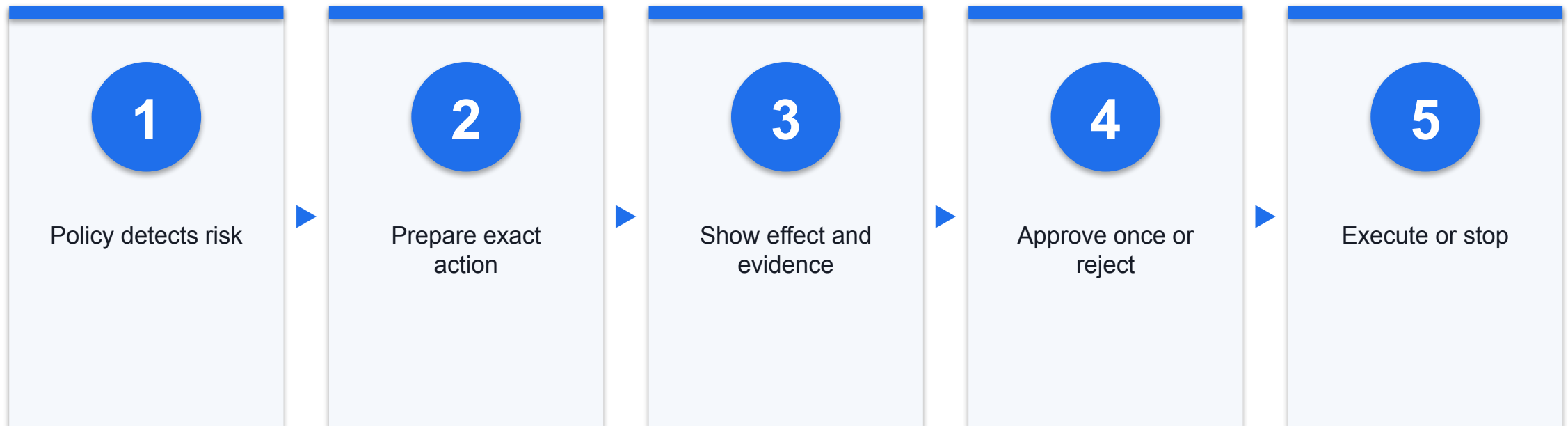
Resources

CPU, memory, time and process limits

Step 6 — Control Network Egress



Step 7 — Deterministic Human Approval



Make Human Approval Meaningful

- Weak approval
- Vague 'continue?'
- Standing permission
- No outcome review

- Meaningful approval
- Exact recipient/data/value
- One action with expiry
- Rejection and override audited

Step 8 — Log the Decision and the Action

Evidence	Why keep it
Trace and identity	Attribute one end-to-end run
Inputs/source provenance	Explain what influenced the model
Policy and approval	Prove control outcome
Tool arguments/result	Reconstruct actual effect

Evidence: SYN · Sources: S10, S14

Step 9 — Monitor Sequences, Not Only Events

1

Sequence

Unexpected tool order or fan-out

2

Boundary

Repeated denials, scope errors or new destinations

3

Consumption

Runaway turns, time, tokens or spend

Step 10 — Red-Team Before Go-Live

Test family	Evidence
Clean tasks	Utility and false blocks
Injection/jailbreak	Attack success by family
Tools/identity	Unsafe or denied action attempts
Memory/supply chain	Persistence and provenance controls
Recovery	Kill switch, revoke and restore

Evidence: DEF · Sources: S09, S12, S14

A Measurable Go-Live Gate

1 Performance

Task success on representative clean cases

2 Security

Attack and unsafe-action results by threat family

3 Operations

Approval quality, detection and recovery tests

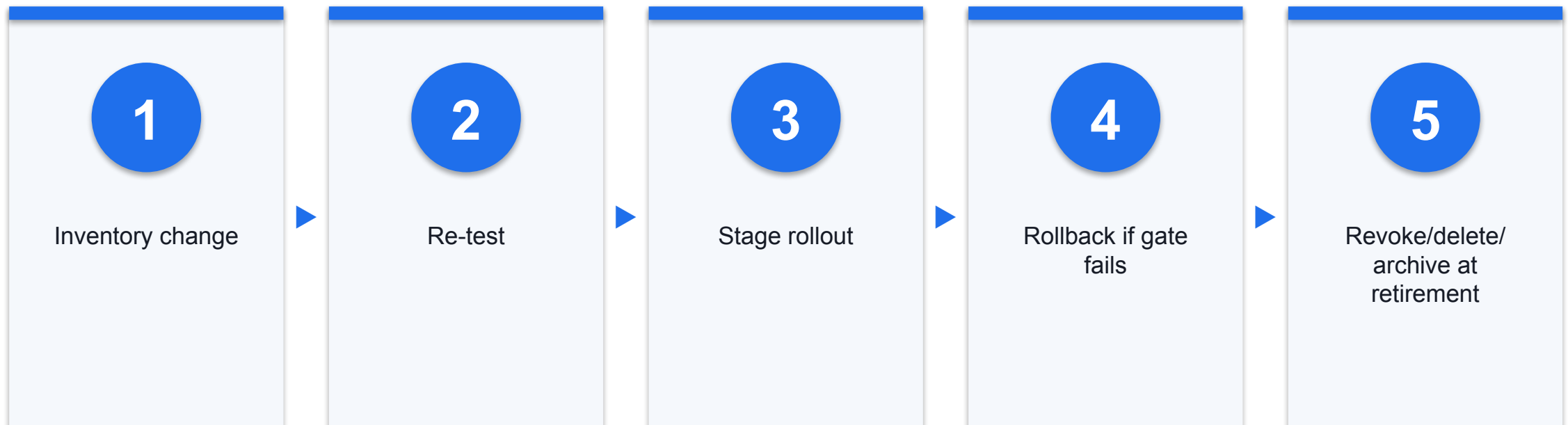
Thresholds are organisation-defined; do not invent universal percentages.

Incident Response for an AI Agent

Phase	Immediate action	Evidence or decision
Identify	Stop work and isolate the runtime	Affected goal, session, tools and data
Contain	Revoke credentials and block egress	Identity, scopes, destinations and timestamps
Assess	Preserve logs and determine impact	Personal data, actions, reversibility and notification test
Recover	Restore clean state and re-test controls	Root cause, corrective action, owner and approval

Evidence: SYN · Sources: S13, S16

Safe Change, Rollback, and Decommissioning



Shared Responsibility Across the Agent Lifecycle

Role	Evidence owned
Provider	Limits, safety data, terms and change notice
Builder	Architecture, tools, provenance and tests
Deployer	Purpose, access, autonomy and approval
Operator	Monitoring, incident response and re-test
User/approver	Verification and escalation

Evidence: SYN · Sources: S14, S15

Responsible AI Principles Become Controls

Principle	Observable control
Accountability	Named owner and residual-risk sign-off
Transparency	Capability, data and appeal notice
Fairness	Segment testing and remediation
Privacy	Minimisation, purpose and retention
Human agency	Meaningful review and remedy

Evidence: SYN · Sources: S09, S14, S15

Activity 5 — Governance Deployment Gate

Realistic synthetic classroom simulation

Review a fictional bank collections agent · Use only synthetic segment and test data · Decide go, conditional go or no-go

1

Scope

2

Map

3

Test

4

Control

5

Evidence

DISCUSSION QUESTIONS

- Review a fictional bank collections agent
- Use only synthetic segment and test data
- Decide go, conditional go or no-go

YOU'LL PRODUCE

A documented risk decision with evidence.

Assesses A1, A2 · Activity pack: [activities/activity-5-agent-governance-deployment-gate/](#) · Full step-by-step facilitation detail: [Learner Guide](#)

Evidence: SIM

Activity 5 — Bias, Privacy, and Autonomy Evidence

Realistic synthetic classroom simulation

Compare performance by affected group · Map personal-data purpose and disclosure · Set read, recommend, prepare and execute boundaries

1

Scope

2

Map

3

Test

4

Control

5

Evidence

DISCUSSION QUESTIONS

- Compare performance by affected group
- Map personal-data purpose and disclosure
- Set read, recommend, prepare and execute boundaries

YOU'LL PRODUCE

A documented risk decision with evidence.

Assesses A1, A2 · Activity pack: [activities/activity-5-agent-governance-deployment-gate/](#) · Full step-by-step facilitation detail: [Learner Guide](#)

Evidence: SIM

Activity 5 — Go, Conditional Go, or No-Go

Realistic synthetic classroom simulation

State evidence and uncertainty · Name conditions, accountable owner and review date · Define kill switch, rollback and escalation

1

Scope

**2**

Map

**3**

Test

**4**

Control

**5**

Evidence

DISCUSSION QUESTIONS

- State evidence and uncertainty
- Name conditions, accountable owner and review date
- Define kill switch, rollback and escalation

YOU'LL PRODUCE

A documented risk decision with evidence.

Assesses A1, A2 · Activity pack: [activities/activity-5-agent-governance-deployment-gate/](#) · Full step-by-step facilitation detail: [Learner Guide](#)

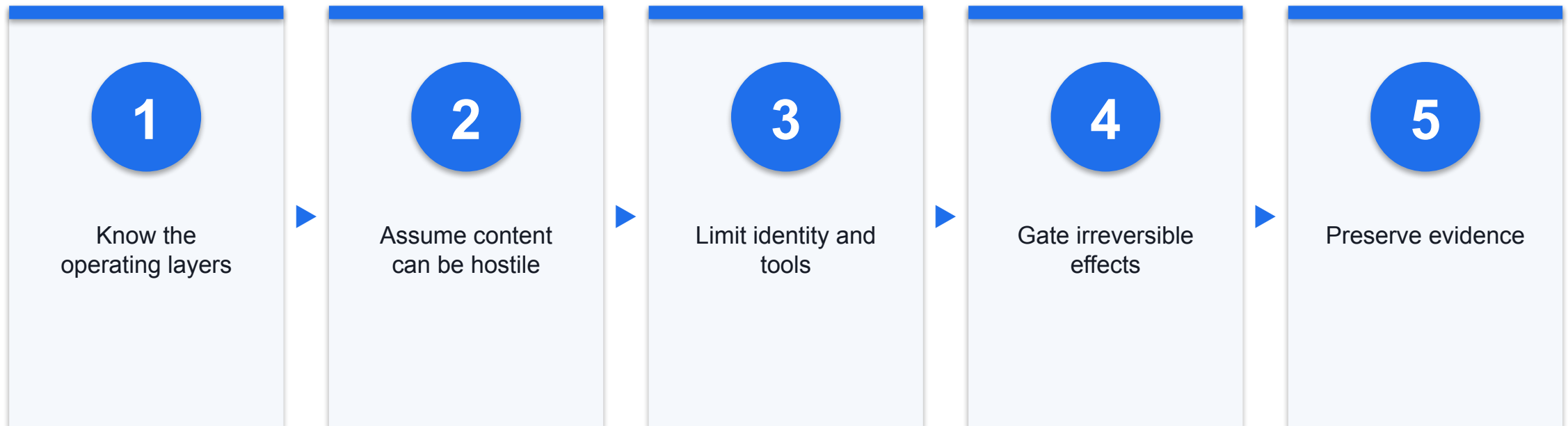
Evidence: SIM

Production Security Checklist

Gate	Minimum evidence
Own	Purpose, tier and accountable roles
Bound	Data, identity, tools, network and autonomy
Assure	Clean/adversarial tests and approval quality
Operate	Logs, alerts, response and rollback
Sustain	Change control and decommissioning

Evidence: SYN · Sources: S10, S12, S14, S15

Course Synthesis — Bound the Authority



Evidence: SYN · Sources: S09, S12, S14

Day 2 Recap — From Attack to Assurance

1

Attack

Trace content through authority to effect

2

Defend

Use layered, deterministic boundaries

3

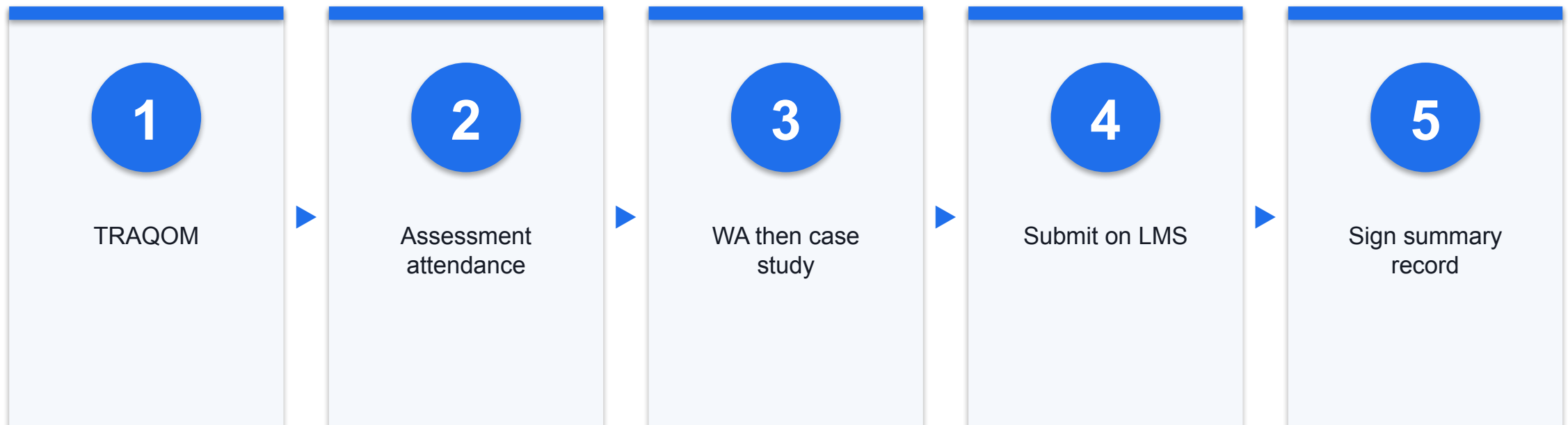
Govern

Measure, own and improve residual risk

Assessment Reminder

- Complete required digital attendance
- Use slides, Learner Guide and activity evidence
- Submit the required assessment files on the LMS

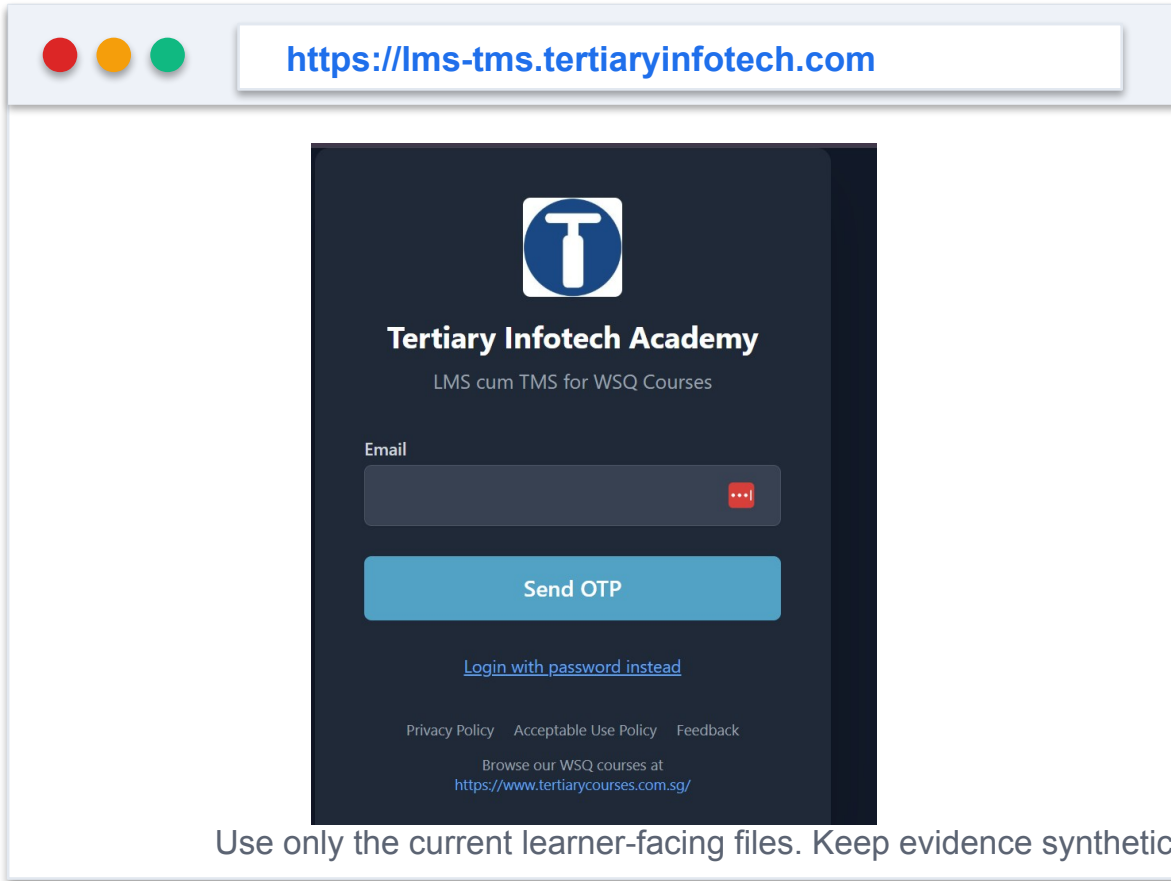
Assessment Flow



Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer or administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.
- Complete the TRAQOM survey at the end of the course — it is required for funding.

Courseware and Assessment on the LMS



The screenshot shows a web browser window with the address bar displaying <https://lms-tms.tertiaryinfotech.com>. The page content is a dark-themed login form for the Tertiary Infotech Academy. At the top is the academy's logo, a white 'T' inside a blue circle. Below the logo, the text 'Tertiary Infotech Academy' is displayed, followed by 'LMS cum TMS for WSQ Courses'. There is an 'Email' input field with a red eye icon for toggling visibility. Below the input field is a blue 'Send OTP' button. A link 'Login with password instead' is positioned below the button. At the bottom of the form, there are links for 'Privacy Policy', 'Acceptable Use Policy', and 'Feedback', followed by the text 'Browse our WSQ courses at <https://www.tertiarycourses.com.sg/>'.

Use only the current learner-facing files. Keep evidence synthetic, local and reversible; never upload secrets or answer keys.

1

Step 1

Download the current learner-facing materials

2

Step 2

Upload only requested assessment evidence

3

Step 3

Do not upload answer keys or production secrets

Thank You — Bound the Authority, Preserve the Evidence

Bound the authority. Preserve the evidence.

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